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CONCEPTUAL SPACES:
A GEOMETRIC FRAMEWORK FOR KNOWLEDGE REPRESENTATION

Abstract

This paper provides a self-contained exposition of Peter Gärdenfors's *conceptual spaces* framework: a geometric middle ground between symbolic logic and connectionist networks in which concepts are convex regions of a quality-dimension space, similarity reduces to distance, and prototype effects follow as theorems. We extend the framework with hierarchical dimensions, Gromov-Wasserstein alignment, category-theoretic foundations, and concrete applications in document processing, personal context modelling, and computing-system design.

Contents

1 Introduction

The world is not made of atoms. It is made of stories.

Muriel Rukeyser

How do minds represent meaning? This question sits at the intersection of cognitive science, linguistics, and artificial intelligence. Classical approaches to knowledge representation have leaned heavily on symbolic logic: concepts are defined by necessary and sufficient conditions, and reasoning proceeds by syntactic manipulation of symbols. While powerful, this approach struggles to account for the graded, context-sensitive, and embodied character of human conceptualisation.

Peter Gärdenfors proposed *conceptual spaces* as a middle ground between the symbolic paradigm and connectionist (neural-network) approaches [Gärdenfors, 2000]. The central intuition is that concepts are best understood not as sets of features or patterns of activation, but as *regions in a geometric space* whose axes correspond to cognitively meaningful quality dimensions such as colour, pitch, temperature, or social dominance.

This geometric stance offers several advantages. Similarity becomes distance; typicality becomes proximity to a prototype; category boundaries emerge naturally as convex regions; and conceptual combination can be modelled by spatial operations. The framework is not merely descriptive: it provides testable predictions about categorisation behaviour, supports compositional semantics, and has been applied to robotics, natural language processing, and knowledge-graph reasoning.

The present paper provides a self-contained exposition of the conceptual-spaces framework. Chapter 2 introduces the formal notion of a conceptual space. Chapter 3 examines the nature and structure of quality dimensions. Chapter 4 develops the geometric theory of natural concepts as convex regions and the role of prototypes. Chapter 5 analyses similarity measures and their relationship to spatial distance. Chapter 6 surveys current applications in cognitive science and AI. Chapter 7 offers conclusions and directions for future work.

2 Foundations of Conceptual Spaces

2.1 Three Levels of Cognitive Representation

Gärdenfors distinguishes three levels at which cognitive agents can be described:

Symbolic level Representations are discrete symbols manipulated by formal rules. This is the level of classical AI and formal logic.

Conceptual (geometric) level Representations are points or regions in a metric space. Inference is driven by geometric operations such as distance and convexity.

Sub-conceptual (connectionist) level Representations are distributed activation patterns in neural networks, with no explicit symbolic or geometric structure.

The conceptual level occupies a privileged position: it is abstract enough to support compositional reasoning yet close enough to perception to explain graded categorisation.

2.2 Formal Definition

Definition 2.1 (Conceptual Space)**.** A *conceptual space* is a triple $\mathcal{C} = (D, \{d_i\}_{i \in I}, \delta)$ where

- $D = D_1 \times D_2 \times \cdots \times D_n$ is a Cartesian product of *quality domains* D_i;
- each domain D_i carries its own metric $d_i\colon D_i \times D_i \to \mathbb{R}_{\geq 0}$ reflecting the internal structure of that domain; and
- $\delta\colon D \times D \to \mathbb{R}_{\geq 0}$ is an overall *distance function* on D, typically a weighted Minkowski metric combining the domain-level distances.

A point $p \in D$ is called a *stimulus point*; it represents the complete description of a particular object or event across all quality dimensions.

2.3 *Weighted Minkowski Distance*

The most common choice for δ is the weighted Minkowski metric:

$$\delta_r(p,q) = \left(\sum_{i=1}^{n} w_i \, d_i(p_i, q_i)^r \right)^{1/r}, \qquad r \geq 1, \tag{2.1}$$

where $w_i > 0$ are salience weights. The Euclidean case ($r = 2$) is appropriate when dimensions are integral (processed holistically), while the city-block metric ($r = 1$) is appropriate for separable dimensions [Shepard, 1987].

The canonical two-level metric structure. Bechberger and Kühnberger [2017] formalise the metric implicit in Gärdenfors's framework into a canonical two-level form that makes the integral/separable distinction structural rather than parametric. Within each quality domain D_k, distances are computed by a weighted Euclidean metric:

$$d_k(p_k, q_k) = \sqrt{\sum_{j \in D_k} w_{kj} \, (p_{kj} - q_{kj})^2}, \tag{2.2}$$

with dimension-level salience weights $w_{kj} > 0$. Across domains, these intra-domain distances are combined by a weighted Manhattan (city-block) sum:

$$\delta(p,q) = \sum_{k=1}^{m} W_k \, d_k(p_k, q_k), \tag{2.3}$$

with inter-domain salience weights $W_k > 0$. This two-level decomposition is not notational convenience: the Euclidean intra-domain metric reflects the holistic, non-decomposable processing of integral dimensions (hue and brightness are inseparable); the Manhattan inter-domain combination reflects the independent, parallel processing of separable domain clusters. The choice of r in (2.1) interpolates between these regimes, with $r = 1$ exactly recovering (2.3).

The two-level structure has a structural consequence for category geometry: under the Manhattan inter-domain combination, globally convex regions are necessarily axis-aligned hyperrectangles in domain-coordinate space, erasing all inter-domain correlations. This observation, which Bechberger and Kühnberger [2017] demonstrate via the age-height covariation example for human children, motivates the replacement of global convexity with star-shapedness in section 4.5. The salience weights at both levels are context-sensitive: varying w_{kj} sharpens attention within a domain, varying W_k reprioritises whole domain clusters. This weight adaptivity is the geometric analogue of context-dependent salience (section 5.6): each constrained projector P_i (section 2.8) implicitly induces a specific weight configuration (w_{kj}, W_k) over the quality dimensions, making the metric operator-conditioned rather than globally fixed.

2.4 Betweenness and Paths

A key structural notion is *betweenness*: point b is between a and c if $\delta(a,b)+\delta(b,c) = \delta(a,c)$. In Euclidean space this reduces to collinearity, and the set of all points between a and c forms the line segment $[a,c]$. Betweenness underlies the definition of convexity used in chapter 4 and supports interpolative reasoning ("something between red and orange is reddish-orange").

2.5 Category-Theoretic Perspective

An orthogonal foundation for conceptual spaces is provided by *applied category theory*. Rather than fixing a single ambient metric space, the category-theoretic viewpoint treats conceptual spaces as objects in a category whose morphisms are *structure-preserving maps*—analogous to functors between categories of metric spaces.

Tai-Danae Bradley's programme at the intersection of category theory, quantum probability, and language modelling provides one bridge: the *functor* from a syntactic category of linguistic expressions to a semantic category of geometric spaces respects compositional structure, so complex meanings are built from primitive ones without losing geometric coherence [Bradley, 2022]. This perspective makes the *naturality* of semantic composition a theorem rather than an assumption.

A complementary line is topos-theoretic: Joshua Wrigley and others study presheaf categories over logical sites as models for structured knowledge spaces, where the Grothendieck topology encodes which covers of a concept are *admissible* [Wrigley, 2023]. This framework naturally accommodates context dependence—different covers model different viewpoints on the same conceptual region—and provides an internal logic for reasoning about concept membership.

Remark 2.2. The category-theoretic and metric-geometric perspectives are complementary rather than competing. Metric structure determines the *local* geometry of a conceptual space (distances, prototypes, convexity), while the categorical structure determines the *global* relationships between spaces (functorial mappings, natural transformations, limits and colimits as conceptual combination operators).

2.6 Typed Sparse Vector Spaces

A MORE RADICAL DEPARTURE from the product-space picture emerges when the basis vectors themselves are *typed* and the product structure is relaxed to a *sparse associative algebra*. Rather than fixing a universal space $D_1 \times \cdots \times D_n$, a *typed sparse vector space* lets each document or concept carry only the quality dimensions it actually uses:

$$\mathbf{v} = \bigoplus_{k \in K} v_k \, \mathbf{e}_k, \qquad K \subseteq \mathsf{Keys}, \quad |K| \ll |\mathsf{Keys}|, \tag{2.4}$$

where each basis element carries an explicit *type annotation*:

$$\mathbf{e}_k = (\mathsf{key}_k,\ \mathsf{type}_k), \quad \mathsf{type}_k \in \{\text{scalar, text, formula, pointer, closure, process-tag, theorem-object, } \ldots\}. \tag{2.5}$$

This is not a classical $\mathbb{R}^n$: it is closer to a *dependent typed record* or *sparse associative algebra*—related to affine geometry and graph/path algebra rather than linear algebra [Deza and Deza, 2009].

Remark 2.3 (Relation to the product-space definition). The typed sparse structure does not contradict definition 2.1; it is a realisation of that definition over an *open-ended key universe* Keys. The full product space $\prod_{k\in\mathsf{Keys}} V_k$ is the ambient space; every concept vector is a point in this space with coordinates outside K absent (set to a distinguished null value). Definition 2.1 fixes a *closed* n-dimensional universe; the typed sparse space opens it to an unbounded universe from which each concept selects its active subset. The distance function δ of definition 2.1 specialises to the shared-key similarity of eq. (2.7): only dimensions shared between two concepts contribute to their distance, which is exactly right when inhabited dimensions are much fewer than the ambient universe. The two structures are therefore related as a fixed-dimensional cross-section is related to the ambient space in which it embeds.

Affine interpretation. Documents are not free vectors but *bound vectors* anchored at an origin, in the spirit of projective geometric algebra (Hestenes). A document represents the path from a root node to a leaf in the knowledge graph, and the difference of two documents represents the directed path between them:

$$\mathbf{v} = \text{path}(\mathsf{root} \to \mathsf{node}), \qquad \mathbf{u} - \mathbf{v} = \text{path}(\mathsf{node}_v \to \mathsf{node}_u). \tag{2.6}$$

This interpretation aligns naturally with citation graphs, ontology traversal, and theorem dependency chains: subtracting two concept vectors yields the *conceptual displacement* between them.

Sparse-key similarity. Because dimensions are heterogeneous, similarity acts only on the *shared keys* of two vectors, with the per-key similarity function dispatched by type:

$$\text{sim}(\mathbf{u}, \mathbf{v}) = \sum_{k\,\in\,\mathsf{keys}(\mathbf{u})\,\cap\,\mathsf{keys}(\mathbf{v})} \text{sim}_k(u_k,\ v_k). \tag{2.7}$$

Representative per-type similarity rules are given in section 2.6.

Nilpotent citation dimensions. Among the typed basis elements of (2.5), citation dimensions occupy a distinguished role. For each citable work j, the basis vector $\mathbf{e}_j^{\text{cite}}$ is the *citation dimension*: a document d carries coordinate $v_j^{\text{cite}} = 1$ if it cites work j, and 0 otherwise. Repeated citations to the same work leave this coordinate unchanged—a second citation to j adds no new information to the document's citation profile. This *idempotency* is captured by treating citation-dimension basis vectors as generators of an exterior subalgebra:

$$\mathbf{e}_j^{\text{cite}} \wedge \mathbf{e}_j^{\text{cite}} = \mathbf{0}. \tag{2.8}$$

Basis type	Similarity rule
Scalar	Numerical kernel (RBF)
Short string	Edit (Levenshtein) distance
Paragraph	Cosine of dense embedding
Formula	Symbolic normalisation + structural match
Pointer	Similarity of target concept
Closure	Evaluation under shared context
Process tag	Categorical (Jaccard) overlap
Theorem object	Proof compatibility check

Table 2.1: Per-type similarity rules for typed sparse basis vectors.

The anti-symmetry of the wedge product—$\mathbf{e}_i^{\text{cite}} \wedge \mathbf{e}_j^{\text{cite}} = -\mathbf{e}_j^{\text{cite}} \wedge \mathbf{e}_i^{\text{cite}}$—further reflects the asymmetry of sequential citation chains: citing A before reasoning toward B is logically distinct from the reverse order in a derivation. Nilpotency (2.8) eliminates citation redundancy; anti-symmetry records reasoning order. The wedge product operates here on the *citation-type basis vectors* within the typed sparse space, not on document vectors themselves—a citation is a typed coordinate, not a citation-as-relation between two document points.

2.7 Three Enrichment Layers

An adequate sparse representation of a scientific document requires three simultaneous semantic layers whose interplay determines the *full meaning* of each basis vector.

Each layer answers a different question: *where?* (A), *what act?* (B), *what proof?* (C).

Layer A — Conceptual / Thesaurus Geometry. Encodes *where concepts live*: broader/narrower relations, ontological neighbourhood, and conceptual distance in quality-dimension space. Layer A defines similarity between basis vectors themselves, not only between documents.

Layer B — Process / Illocutionary Semantics. Encodes *what logical act is being performed*. Representative process tags for scientific prose:

Tag	Meaning
`DEFINITION`	Introduces a new concept
`DERIVATION`	Proves from prior results
`REFERENCE_USE`	Applies a result verbatim
`REFERENCE_EXTEND`	Generalises or adapts a result
`CONJECTURE`	Asserts without proof
`ANALOGY`	Maps structure across domains
`EXPERIMENT`	Grounds a claim empirically
Citation recontextualisation subtypes	
`CIT_AXIOM`	Cited result treated as unchangeable premise
`CIT_METHOD`	Technique borrowed and adapted to new context
`CIT_CONTRAST`	Explicit disagreement or contradiction
`CIT_RECONTEXT`	Framework extended to a domain not covered by source

Layer C — Type-Theoretic / Lean Semantics. Maps scientific acts to formal logical structure using Lean 4 type theory [de Moura and Ullrich, 2021] as grounding:

Scientific act	Lean 4 interpretation
Definition	Type introduction
Derivation	Proof construction
Citation	Lemma application
Conjecture	Unresolved type (`sorry`)
Derivation gap	Metavariable

Citation recontextualisation as a semantic layer. The four citation subtypes above are not merely bibliographic metadata; they carry distinct geometric meanings in quality-dimension space. A `CIT_AXIOM` citation treats the cited result as a fixed point: the citing document's conceptual position is constrained to lie within the star-shaped region of the cited work (section 4.5). A `CIT_METHOD` citation introduces a *process morphism*: the technique is transported across domains while its logical skeleton is preserved. A `CIT_CONTRAST` citation asserts an *antipodal displacement*: the citing work occupies a region separated from the cited work by at least one quality dimension. The most geometrically significant modality is `CIT_RECONTEXT`: if work P_i cites work P_j with tag `CIT_RECONTEXT`, then the conceptual displacement $\overrightarrow{P_j P_i}$ is non-zero in at least one quality dimension k where the cited work had zero projection,

$$\exists\, k\colon\, \pi_k(P_i) \not\subseteq \pi_k(P_j), \tag{2.9}$$

formally capturing creative reuse as a domain-extension vector. The hidden-novelty conjecture of the following paragraph is operationalised precisely by this condition: a `CIT_RECONTEXT` citation that is tagged only as `REFERENCE_USE` in the surface text constitutes a hidden novelty signal detectable by the Layer B/C mismatch mechanism.

Scientific understatement detection (conjecture). Scientific writing routinely *understates* novelty behind neutral citation language: the phrase "using Smith et al." may conceal a generalisation to a new domain, an implicit contradiction, or an unstated theorem extension. The three-layer framework makes this discrepancy explicit: a mismatch between the stated process tag (Layer B) and the reconstructed logical structure (Layer C) is a candidate *hidden novelty signal*. We conjecture that measuring this discrepancy across a citation network would provide an automated proxy for the scientific significance of unacknowledged contributions—a research programme that the framework makes tractable but which remains to be validated empirically.

Remark 2.4 (Composition rule and epistemic status). Open Question 7 of chapter 7 explicitly acknowledges that the rule by which Layers A, B, C interact—summed, weighted, or composed via a learned mixing function—is not yet specified. Until such a rule is established, any claim that depends on the *relative weighting* of layers (e.g. the understatement hypothesis above) carries the epistemic status of a research programme, not a validated capability. Claims in this manuscript that invoke the three-layer framework as an explanatory substrate should be read accordingly.

Remark 2.5. The typed sparse framework and the metric product space of definition 2.1 are complementary rather than competing: metric geometry captures *where* a concept sits in quality-dimension space, while the three enrichment layers capture *how it arrived there* (derivation), *what kind of object it is* (type), and *what it formally proves* (proof term). A full knowledge-representation system requires both.

Remark 2.6 (Proof digestion as a fourth semantic layer, after Tao). Terence Tao's analysis of AI-assisted mathematics [Tao, 2026] identifies a systematic gap between the *explicit* goal of mathematical work (prove theorem, verify bound) and its *implicit* goals: exposition, modularity, teachability, attribution, insight transfer, and the training of junior researchers. Automated systems excel at explicit goals while leaving implicit goals unmet—a decoupling that produces correct but opaque proofs, the new bottleneck Tao terms *digestion*.

The three enrichment layers A, B, C partially address implicit goals: Layer A encodes conceptual neighbourhood (reusability context); Layer B tags illocutionary acts (including pedagogical acts such as `ANALOGY`); Layer C provides the formal proof term (teachability substrate). However, none explicitly *optimises* for digestion. We therefore propose a **Digestion Layer D** whose metric is the *Semantic Digestion Score*:

$$\mathrm{SDS}(c) \;=\; w_1\,\mathrm{compress}(c) + w_2\,\mathrm{navigate}(c) + w_3\,\mathrm{teach}(c) + w_4\,\mathrm{reuse}(c), \tag{2.10}$$

where $\mathrm{compress}(c)$ is the inverse size of the minimal tiddler set that captures concept c without loss; $\mathrm{navigate}(c)$ is the inverse of the average shortest-path length from c to all related concepts in the CSP graph; $\mathrm{teach}(c)$ is an indicator for the presence of a pedagogical projection (spoken-math form, slide deck, or interactive tutorial); and $\mathrm{reuse}(c)$ is the normalised in-degree of c in the derivation DAG (`u:` field of section 6.26). High SDS identifies concepts that are simultaneously compact, well-connected, pedagogically accessible, and widely reused —Tao's criterion that "someone can give a talk" becomes $\mathrm{teach}(c) = 1$.

Digestion is not post-hoc: every CSP compilation pass should be evaluated on SDS alongside correctness, making it an integral objective of the semantic compiler (section 6.11).

2.8 Constrained Semantic Projection Operators

An LLM prompted with different instructions does not merely retrieve different information—it operates as a *different semantic projector* onto a different latent subspace. Let D denote the space of all raw documents and $\{S_i\}$ a family of projection-specific latent subspaces (grammatical, rhetorical, ontological, proof-structural, citation-network, and so on). A *constrained semantic projection operator* is:

$$P_i\colon D \to S_i, \qquad P_i = \mathcal{L}(\,\cdot\,;\, \mathrm{prompt}_i), \tag{2.11}$$

where $\mathcal{L}(\,\cdot\,;\,\mathrm{prompt}_i)$ denotes the language model conditioned on prompt prompt_i. Each prompt selects a semantic projection; different prompts induce different coordinate systems on the same source document.

Ablative prompting as coordinate isolation. An *ablative* prompt instructs the model to attend exclusively to one semantic register: "Analyse ONLY grammar; ignore all content and meaning" forces P_i to project onto the grammatical subspace S_{gram}; "Extract ONLY citation structure; ignore prose" forces projection onto S_{cite}. An unconstrained prompt entangles all subspaces simultaneously—syntax, discourse, ontology, rhetoric, and scientific interpretation collapse into a single undifferentiated activation pattern. This is the LLM analogue of the *single-space fallacy* of section 2.15: imposing one metric on all semantic types at once.

The projection operators connect directly to the three enrichment layers of section 2.7: Layer A projectors target ontological neighbourhood; Layer B projectors isolate illocutionary acts (definition, derivation, conjecture); Layer C projectors recover type-theoretic structure. Running all three in parallel and combining their outputs reconstructs the full enriched typed-sparse representation without conflation.

Stable symbolic anchors. Factorised projection requires identifiers that remain stable across all subspaces. A *stable symbolic anchor* is a persistent semantic ID assigned to a document element before any projector fires:

$$\texttt{\{\{CSPPaper.equation.0042\}\}} \;\mapsto\; F \in \mathcal{M}_{\mathrm{formula}}, \tag{2.12}$$

so that the inline mention `{{CSPPaper.inline.0017}}` and the display equation `{{CSPPaper.equation.0042}}` refer to the same formula object F regardless of which projector is active. This is the operational realisation of the transclusion principle of section 6.25: one semantic object, arbitrarily many views, zero duplication.

Latent formal structure from prose. Constrained projection can *recover* latent formal structure that the author never stated explicitly. A description such as "observable duals and admissible probes" contains sufficient semantic content for a type-theoretically constrained projector P_C (Layer C) to reconstruct:

$$O(X,Y) = \bigl\{ f\colon X \to Y \;\big|\; f \text{ admissible} \bigr\}, \tag{2.13}$$

without the author having written this type signature. The projector acts as a *type-inference engine*: it checks whether the prose is consistent with a formal type and synthesises the signature as a Layer C annotation when consistent. This capability directly supports the scientific-understatement detection conjecture of section 2.7: hidden novelty corresponds to a Layer B tag (e.g. `REFERENCE_USE`) that is inconsistent with the Layer C structure P_C recovers.

2.9 Formula Ontology: Three Representation Layers

A common source of confusion in knowledge representation is treating syntactic tokens as conceptual objects. A mathematical formula such as $F = ma$ exists simultaneously at three distinct layers that must be kept separate [Kolbe, 2025]:

Referential layer Physical or mathematical *entities* and their relations (force, mass, acceleration; the proportionality relation). These exist independently of any formula notation.

Semantic layer The *conceptual relation* with its modality: a definition, an empirical law, an approximation. "Force equals mass times acceleration" is a law-like claim, not a definition.

Syntactic layer The *serialisation* into a particular notation: LaTeX string `F = m a`, spoken text "eff equals em times ay", or a computer algebra system expression. This is a *view* of the semantic layer, not the layer itself.

In the typed sparse framework, a formula object carries all three layers: the syntactic `latex` key, a `relation` key encoding the conceptual structure, and `symbol` keys linking to entity definitions. An LLM reading only the syntactic layer is forced to guess the semantic and referential content—precisely the failure mode that motivates the CSP approach.

2.10 The Formula as Manifold: Projection Geometry of Mathematical Expressions

The three-layer analysis of the preceding section reveals a deeper structural claim: *the equation is not the object.* The LaTeX string $F = ma$, the Lean proof term, the spoken utterance "eff equals em times ay", and the computer-algebra AST are not the formula—they are *coordinate charts* on a single underlying object that inhabits none of them fully. The appropriate mathematical framework is not a document database or a vector table but a *manifold with projection morphisms.*

The formula manifold. Let $\mathcal{M}_{\text{formula}}$ denote the *formula conceptual space*: a high-dimensional space whose points represent mathematical expressions as abstract objects, independent of any particular notation. A single formula $F \in \mathcal{M}_{\text{formula}}$ is a tuple of components spanning all available coordinate systems:

$$F = \big(v_{\text{latex}},\ v_{\text{lean}},\ v_{\text{cas}},\ v_{\text{semantic}},\ v_{\text{visual}},\ v_{\text{historical}},\ v_{\text{citation}},\ v_{\text{proof}},\ v_{\text{embedding}},\ \ldots\big), \tag{2.14}$$

where each component v_α lives in a different coordinate space $\mathcal{R}_\alpha$ appropriate to that representation. Projection morphisms

$$\pi_\alpha \colon \mathcal{M}_{\text{formula}} \ \to\ \mathcal{R}_\alpha \tag{2.15}$$

are the *coordinate charts*: they render the abstract formula object into a specific representational format. No single chart is the formula itself—the formula is the invariant object that all charts describe.

Projection π_α	**Target space** $\mathcal{R}_\alpha$	**Metric / structure**
π_{latex}	LaTeX string	Edit distance
π_{katex}	Rendered HTML	Pixel/layout similarity
π_{lean}	Lean 4 proof term	Proof-term equivalence
π_{cas}	CAS operator-tree (AST)	Tree edit distance
π_{semantic}	Thesaurus neighbourhood	Graph diffusion
$\pi_{\text{embedding}}$	Latent vector	Cosine / Euclidean
π_{citation}	Citation graph node	Diffusion kernel
π_{proof}	Theorem dependency DAG	Overlap / containment
π_{visual}	Image (MathPix region)	Perceptual distance
π_{spoken}	Phoneme graph	Speech edit distance
π_{atlas}	"Mathematistan" coordinates	Geographic proximity
π_{pedagogy}	Explanation graph (YouTube, StackEx.)	Semantic cloud
$\pi_{\text{transcl.}}$	TiddlyWiki hypertext neighbourhood	Backlink topology

Table 2.2: A partial atlas of projection charts on $\mathcal{M}_{\text{formula}}$. Each row is a coordinate chart π_α; no single chart is privileged.

The Nash embedding analogy. Nash's isometric embedding theorem guarantees that every smooth Riemannian manifold M^n admits an isometric embedding into $\mathbb{R}^N$ for sufficiently large N. The key lesson is that a curved manifold cannot always be faithfully represented in a low-dimensional flat space: the intrinsic curvature of M^n requires the additional dimensions of $\mathbb{R}^N$ to express without distortion. The same principle applies to $\mathcal{M}_{\text{formula}}$: the "true" relational structure of a formula — its proof dependencies, its semantic neighbourhood, its historical lineage, its typographic identity—cannot be faithfully encoded by any single projection π_α alone, however expressive. This is precisely why AI systems that represent mathematical knowledge through token embeddings alone collapse the manifold into a flat low-dimensional subspace, destroying the curvature that carries relational meaning. A faithful representation requires multiple charts, held simultaneously, with explicit inter-chart consistency conditions (the sheaf gluing of section 2.15).

Projection composition algebra. Since each projection π_α is a morphism in the enriched category **CSP** (section 2.15), projections can be *composed*:

$$\pi_\beta \circ \pi_\alpha \colon \mathcal{M}_{\text{formula}} \xrightarrow{\pi_\alpha} \mathcal{R}_\alpha \xrightarrow{\pi_\beta} \mathcal{R}_\beta. \tag{2.16}$$

Examples: $\pi_{\text{katex}} \circ \pi_{\text{latex}} \circ \pi_{\text{normalised AST}}$ renders a canonical form from the operator tree through LaTeX into displayed HTML; $\pi_{\text{embedding}} \circ \pi_{\text{proof}}$ embeds the proof-dependency subgraph into latent space. Composition may fail to commute (rendering before normalisation yields a different result than normalising before rendering), making the diagram of projections a non-commutative object—an important constraint when building retrieval pipelines that traverse multiple charts.

The non-commutativity of projection composition is not a defect. It reflects genuine semantic information: the path through which you arrive at a representation matters for what structural information is preserved.

Multi-composite embeddings. A single vector embedding of F is one value of $\pi_{\text{embedding}}$, computed from whichever input the encoder processes. A *multi-composite embedding* stacks type-specific embeddings from different projections:

$$e(F) = e_{\text{semantic}}(F) \oplus e_{\text{proof}}(F) \oplus e_{\text{visual}}(F) \oplus e_{\text{historical}}(F), \quad (2.17)$$

where $\oplus$ is direct sum (concatenation with independent metric on each block). This connects directly to Matryoshka Representation Learning (section 6.17): the importance ordering on embedding dimensions corresponds to the ordering of projection blocks by semantic centrality—proof-structure first, historical lineage last. Unlike a single token embedding, eq. (2.17) preserves the type-local metric structure: similarity computed within each block uses the metric appropriate to that projection's space (section 2.10).

Differential geometry of formula space. At each point $F \in \mathcal{M}_{\text{formula}}$, the tangent space $T_F\mathcal{M}_{\text{formula}}$ is the space of *local perturbations*: nearby rewrites, notation variants, limit cases, and symbolic substitutions that leave the semantic core of the formula intact. A tangent vector $\mathbf{v} \in T_F\mathcal{M}$ represents a direction of infinitesimal change—"replace sin with its small-angle approximation θ," "lift from $\mathbb{R}$ to $\mathbb{C}$," "pass to the continuum limit." The *curvature* of $\mathcal{M}_{\text{formula}}$ at F measures how quickly these directions diverge: high curvature corresponds to semantic instability (nearby rewrites lead to radically different meaning), ambiguity (many inequivalent directions with similar syntactic distance), or paradigm boundaries (the formula sits at the edge between two theoretical frameworks). In this differential- geometric view, an operator such as ∇ acting on the semantic potential of formula space produces a *semantic flow field*: the gradient points toward the next natural generalisation, divergence measures conceptual fertility (how many new formulas a given result implies), and non-zero curl signals cyclic dependencies or mutually referencing families of results.

Information-geometric projection. When a formula appears in ambiguous notational contexts, its meaning is not a single point in $\mathcal{M}_{\text{formula}}$ but a *probability distribution* over possible interpretations:

$$p(\text{meaning} \mid \text{context}) \in \Delta(\mathcal{M}_{\text{formula}}), \quad (2.18)$$

where Δ denotes the space of probability measures. Ambiguous notation (e.g. $\sin^{-1}$ meaning both arcsin and $1/\sin$) corresponds to high-entropy distributions; canonical, unambiguous theorems correspond to sharp, near-Dirac distributions. This information-geometric view provides a principled retrieval confidence score: a query retrieves the formula whose distribution $p(\text{meaning} \mid \text{query})$ has the highest overlap with the query's own distribution—the CSP generalisation of the confidence $> \theta$ clause in eq. (6.17).

2.11 Geometric Algebra Interpretation

THE AFFINE INTERPRETATION of eq. (2.6) can be systematically extended to *geometric algebra* (GA), providing a rich algebraic structure for operations on conceptual spaces [Deza and Deza, 2009]. Following Harper's *full unified* view—which explicitly distinguishes bound vectors (anchored to a reference) from free vectors (translation-invariant differences)—a corpus-level conceptual space admits the following algebraic objects:

Bound and free vectors. Let O denote a *corpus baseline* (the centroid of the reference corpus). Each paper or concept P_i is represented by a *bound vector* $\overrightarrow{OP_i}$, its full semantic embedding relative to the baseline. The *free vector* (conceptual displacement) between two papers is:

$$\overrightarrow{P_iP_j} = \mathbf{v}_j - \mathbf{v}_i, \tag{2.19}$$

representing the novelty direction, the method change, or the assumption shift that separates the two works.

Wedge product and theory patches. For two free vectors $\mathbf{a}$, $\mathbf{b}$, the *exterior (wedge) product*:

$$\mathbf{a} \wedge \mathbf{b} \tag{2.20}$$

creates a *bivector*—an oriented 2-dimensional conceptual plane. This is the smallest conceptual subspace containing both directions.

$\mathbf{a} \wedge \mathbf{b} = 0$ iff the two papers are conceptually parallel (same direction); $\|\mathbf{a} \wedge \mathbf{b}\|$ large iff they are genuinely independent.

Bivectors represent *local theory patches*: the minimal 2-dimensional manifold of variation between two independent conceptual directions (e.g. the plane spanned by "method change" and "assumption change").

Trivectors and domains. Three independent conceptual directions generate a *trivector* $\mathbf{a} \wedge \mathbf{b} \wedge \mathbf{c}$—a 3-dimensional oriented theory volume. A family of related models differing along three independent axes (method, data, theory) occupies such a volume. The *pseudoscalar*

$$I = \mathbf{e}_1 \wedge \mathbf{e}_2 \wedge \cdots \wedge \mathbf{e}_n \tag{2.21}$$

encodes the full domain of discourse and enables the *duality* operation: for any multivector A, its dual $A^* = AI^{-1}$ converts a conceptual direction into the hyperplane orthogonal to it—a *classifier boundary*.

Graph-CSP connection via Hodge decomposition. When a citation graph is given, a spanning tree provides unique root-to-leaf paths that realise bound vectors, while the *extra edges* (non-tree edges) generate cycles. These cycles are the geometric-algebra analogue of bivectors: they represent *inconsistencies*, *competing theories*, or *mutual dependencies* that cannot be captured by a simple tree embedding.

The *Hodge decomposition* of the citation graph then separates:

- *Gradient part* (exact): the spanning tree, encoding hierarchical derivation.
- *Curl part* (co-exact): cycles, encoding competing or mutually dependent theoretical commitments.

- *Harmonic part*: global topological features of the citation complex (holes in the knowledge structure).

This decomposition makes the algebraic structure of a citation network explicit and computable, providing a principled basis for tasks such as inconsistency detection and knowledge gap localisation.

Versors as context transformations. In geometric algebra, any composition of reflections and rotations—and in conformal GA, also translations and scalings—can be expressed as a *versor*: for a versor V built from unit vectors, every transformation takes the *sandwich form*

$$X' = VXV^{-1}. \tag{2.22}$$

This single equation unifies all grade-preserving transformations: points remain points, lines remain lines, and metric relations are preserved. Applied to conceptual spaces, a *context transformation*—moving from one observer's perspective to another, from one document domain to another, or from one user's prototype coordinates to a shared public space—is not an arbitrary linear map but a versor in the geometric algebra of the quality space. The versor group is closed under composition: the product of two legitimate perspective shifts is itself a legitimate perspective shift. Crucially, versors are *metric-preserving by construction*—each generator is a unit-vector reflection that preserves the quadratic form of quality space, so composing versors produces only transformations that respect the metric structure of the conceptual space. An arbitrary invertible matrix, by contrast, can shear and rescale quality axes, destroying the distance relations that give similarity its meaning. The versor constraint is therefore a structural guarantee that perspective shifts remain within the class of geometry-preserving transformations, making the versor formalism the natural algebraic home for the context transformation morphisms of chapter 6.

2.12 Similarity as Mathematical Foundation

BEFORE QUALITY DIMENSIONS can be defined, there must be a more primitive relation from which they emerge. Rudolf Carnap's *Aufbau* [Carnap, 1928] proposes exactly this: ground all empirical concepts in a single binary relation $\mathrm{Er}(x, y)$ —"x is remembered as similar to y"—and reconstruct quality structure via *quasi-analysis*.

Quasi-analysis and similarity circles. A *similarity circle* is a maximal set of mutually similar objects:

$$\mathcal{S}_x = \{y \mid \mathrm{Er}(x, y)\}. \tag{2.23}$$

Quality dimensions emerge as the axes that best separate these circles. Carnap's programme is therefore *inductive*: structure is discovered from similarity data rather than imposed. The fundamental difficulty—known since Goodman's critique—is that quasi-analysis does not always produce unique decompositions: distinct quality dimensions may generate the same pattern of similarity circles.

Mormann's sheaf representation theorem. Thomas Mormann [Mormann, 2009] resolved part of this non-uniqueness by characterising when a similarity space admits a representation as a *bundle of qualities*. Let $f\colon S \to 2^Q$ assign each object its set of qualities. The *extension space*

$$E = \{(s, q) \mid q \in f(s)\} \tag{2.24}$$

carries the structure of an *étale space* (local homeomorphism over S) whenever the similarity relation satisfies the *Similarity Neighbourhood Identity* (SNI) axiom: two objects are similar if and only if they share the same similarity neighbourhood. Under SNI, the étale space is unique up to isomorphism—Mormann's existence and uniqueness theorem for quasi-analyses. Sheaf cohomology over the resulting Alexandrov topology then measures global obstructions to quality reconstruction, connecting Carnap's programme to modern topological data analysis.

SNI is the categorical analogue of the T_0 separation axiom: distinct objects have distinct open neighbourhoods in the induced Alexandrov topology.

Similarity algebra. A complementary algebraic perspective is provided by the *similarity algebra* framework of Ghojogh and Amanpour [Ghojogh and Amanpour, 2026]: classical axioms (associativity, transitivity of equivalence, triangle inequality) are replaced by *approximate* versions holding up to a tolerance $\varepsilon \geq 0$:

$$\mathrm{sim}(a, c) \;\geq\; \mathrm{sim}(a, b) \cdot \mathrm{sim}(b, c) - \varepsilon. \tag{2.25}$$

As $\varepsilon \to 0$ the system collapses to classical structures (ultrametric, equivalence relation, Boolean lattice). The non-zero tolerance regime models *graded similarity*: the algebraic distance to classical structure is itself an information-carrying quantity, recording how far a domain departs from clean categorical boundaries. This is directly relevant to CSP regions that are nearly convex but not exactly so—the normal situation in empirical and perceptual domains.

Connection to homotopy type theory. The *univalence axiom* of homotopy type theory [The Univalent Foundations Program, 2013] asserts that equivalence of types is propositionally identical to identity: $(\mathcal{A} \simeq \mathcal{B}) \simeq (\mathcal{A} = \mathcal{B})$. In the CSP context: two conceptual spaces that are structurally equivalent—sharing the same similarity structure, prototype layout, and convexity geometry—are, for all inferential purposes, the *same* space. This gives a precise sense in which *shape of meaning* is more fundamental than *label of meaning*, and underwrites the CSP programme of treating structural similarity as the primary epistemic resource rather than nominal type membership.

Goodman's non-uniqueness problem and its resolution. Nelson Goodman's critique of quasi-analysis goes deeper than noting that similarity is context-sensitive. His *non-uniqueness argument* shows that a given pattern of similarity circles can, in principle, be explained by many different quality decompositions: the similarity data do not single out a unique set of quality dimensions. Carnap attempted to meet this objection with a *fundiert* (grounded) meta-axiom: among all quasi-analyses compatible with the data, prefer those whose quality classes form well-founded hierarchies without circular dependence. This anticipates modern structural selection criteria (minimality, regularity, Occam penalties), but falls short of a complete resolution.

Mormann's SNI theorem provides the missing ingredient: uniqueness follows not from an ad hoc preference axiom but from a topological separation condition on the similarity structure itself. When SNI holds, the étale space is unique up to isomorphism, and Goodman's underdetermination dissolves—not because we rule out alternatives by fiat but because the topology of similarity circles forces a unique reconstruction. Where SNI fails, the non-uniqueness is genuine and measures a real ambiguity in the domain: multiple orthogonal quality systems are consistent with the observed similarity data, a situation that CSP models as a degenerate manifold (several independent axes with identical projections onto the observation basis).

Mereotopological similarity. A further tradition connects similarity to *part-whole* relations rather than point distances. Mereotopology (region-based spatial logic, e.g. the RCC calculus) takes *contact* and *parthood* as primitives and derives topological concepts (interior, closure, boundary) without presupposing a metric. Two regions are "mereotopologically similar" if they share indistinguishable connectivity patterns— a notion encoded in Boolean algebras with adjacency (ultrafilters over the contact lattice). This extends Carnap's programme to domains where objects are inherently extended and boundaries are more natural than points: anatomical regions, geographic zones, legal territories, software module boundaries. In the CSP setting, mereotopological similarity implies that some quality axes are not distance functions but *part-of hierarchies*, and prototype membership is determined by containment rather than proximity. The resulting category of "mereotopological conceptual spaces" is more expressive than metric spaces and provides the geometric underpinning for the hierarchical quality dimensions introduced in chapter 3.

2.13 The CSP Dual Space as Multi-Scale Probe Algebra

Every vector space carries a dual of linear functionals. For a conceptual space with heterogeneous, possibly fractal quality domains, the classical dual $\mathcal{C}^* = \mathrm{Hom}(\mathcal{C}, \mathbb{R})$ is too restrictive: the notion of "linear" is undefined globally when axes have different geometric characters. A more adequate notion is the *observable dual*:

$$\mathcal{C}^* \;:=\; F(\mathcal{C},\, S), \tag{2.26}$$

where S is a space of measurement values (not necessarily $\mathbb{R}$) and F is a class of *admissible probes*—functions consistent with the local structure of $\mathcal{C}$. For a continuous axis, F contains Lipschitz maps; for a tree-structured axis, hierarchical projection functions; for a fractal axis, wavelet-like scale-localised probes.

Local duals and the fibered structure. When each quality domain D_i has its own dual $D_i^* = F_i(D_i, S_i)$, the global dual is not a Cartesian product but a *fibered integral*:

$$\mathcal{C}^* \sim \int^{\oplus} D_i^* \, d\mu(i), \tag{2.27}$$

where μ is a measure over the index set of quality domains. Probes in this fibered dual can mix scales and axes: a single observable may activate only on a sub-branch of one domain conditioned on the value of another—a feature that neither classical linear duals nor fixed-basis Fourier transforms can express.

This is the direct-integral decomposition of a Hilbert space into fibres, adapted to the non-Hilbert setting of heterogeneous quality domains.

Relation to harmonic analysis. The Fourier transform exploits the Pontryagin dual of the translation group: its basis functions e^{ikx} are characters of a Lie group acting on a homogeneous space. A CSP with curved, hierarchical, or discrete quality domains admits no such global symmetry. The appropriate generalisation replaces Fourier modes with: *wavelets* (scale-localised probes for multi-resolution hierarchies), *graph Fourier modes* (eigenfunctions of the graph Laplacian for citation networks and ontology graphs), and *diffusion eigenfunctions* (scale-dependent probes on non-flat manifolds). In each case the dual is a structured collection of geometry-adapted probes rather than a homogeneous function space.

Holographic sub-spaces. In high-dimensional conceptual spaces information is not uniformly distributed across all quality dimensions. There often exist low-dimensional *boundary subsets* $B \subset \mathcal{C}$ from which the full space can be approximately reconstructed:

$$\text{observables on } B \;\Rightarrow\; \text{reconstruction of } \mathcal{C}. \tag{2.28}$$

Such subsets correspond to high-curvature regions in the concept manifold—semantic pivots, decision boundaries, or prototype cores—that carry disproportionate information. This is the CSP analogue of holographic duality (note: the analogy differs from AdS/CFT physics, where a lower-dimensional boundary encodes a higher-dimensional bulk via quantum gravity; here the "boundary" is not lower-dimensional but information-dense—a high-curvature semantic stratum chosen by information content, not by spacetime geometry): a low-dimensional information-dense stratum of quality-dimension space. In practical terms, it identifies which quality dimensions are most worth measuring when resources are limited: measure at the semantic pivots, and the rest can be reconstructed from them.

Remark 2.7. The CSP dual space, understood as a multi-scale structure-adaptive probe algebra, unifies three classical measurement paradigms: linear duals (all linear observables), Fourier space (probes diagonalising a symmetry group), and holographic encoding (redundant boundary measurements). A conceptual space with heterogeneous quality domains requires all three simultaneously, because its domains are continuous, discrete, hierarchical, and asymmetrically informative at the same time.

2.14 Declarative and Procedural Conceptual Spaces

THE CLASSICAL CSP FRAMEWORK models *objects*: a colour, a shape, a bird, a legal norm occupies a point or region in quality-dimension space. But cognitive agents do not only represent what things *are*; they also represent what to *do*—and these two kinds of representation require orthogonal axes.

Declarative CSP. The standard framework is a *declarative* conceptual space: its quality dimensions describe properties of entities (hue, temperature, social dominance, grammatical category). Concepts are convex regions, prototypes are central points, and similarity is distance. This is the space of nouns and adjectives: *theorem*, *formula*, *document*, *reference*, *entity*, *constraint*.

Procedural CSP. A complementary *procedural* conceptual space has quality dimensions that describe properties of *transformations* rather than objects: branching factor, reversibility, resource cost, commitment depth, escalation level. Points in this space are not entities but *policies* or *skills*: *debugging*, *refactoring*, *negotiation*, *planning*, *challenge-response grilling*. Two skills are similar if they share transformation structure (both involve iterative hypothesis revision with escalating constraints), not if they share object content. Gärdenfors [2024] develops a closely related extension of the CSP framework to events and causal dynamics, showing that event representations require "action spaces" orthogonal to the static quality-dimension spaces of objects—independently arriving at the same architectural separation we propose here.

The "grill-me" skill [Kolbe, 2025] illustrates the distinction precisely. It defines a decision procedure:

$$\text{question} \to \text{confidence estimate} \to \text{contradiction test} \to \text{escalation} \to \text{reformulation} \to \text{edge case},$$

which is a path through a *policy graph*—a directed graph over epistemic states whose edges are query-and-test operations. This structure is invisible in declarative quality-dimension space; it requires procedural axes such as *escalation level* (how challenging the current question is), *commitment depth* (how many prior answers constrain the current one), and *contradiction pressure* (how close the agent is to a logical inconsistency).

Two orthogonal spaces, one architecture. The typed sparse vector space of section 2.6 already separates basis vector types: `theorem`-typed vectors live in the declarative space; `process`-typed vectors (Layer B of section 2.7) live in the procedural space. The enrichment-layer framework therefore implicitly handles both, but only if procedural axes are treated as first-class dimensions rather than metadata tags. A fully realised CSP architecture for agentic systems must:

1. maintain separate quality-dimension bases for declarative and procedural sub-spaces,
2. define similarity metrics appropriate to each (distance for declarative; policy-graph edit distance or bisimulation for procedural), and
3. allow *cross-space projections*: a theorem (declarative) can activate a proof-search skill (procedural) via a typed morphism between the two spaces.

This separation is directly relevant to the multi-agent expert system architecture of section 6.21: each expert's *Skills* component (eq. (6.12)) is a set of procedural CSP points, while its *State* is a declarative CSP patch.

2.15 Type-Local Geometries and the Enriched Category of Concepts

A PERSISTENT TEMPTATION in knowledge-representation engineering is to project all concept types into a single latent vector space and rely on a single distance function. This architecture destroys the properties that make typed conceptual spaces useful: hierarchy collapses to proximity, provenance is lost in averaging, compositionality becomes opaque, and transclusion—the identity-preserving reference of the same object from multiple contexts—is indistinguishable from co-occurrence. The correct conclusion is not to find a *better* single embedding; it is to abandon the single-geometry assumption entirely.

The single-space fallacy. A theorem, a formula, a citation, a diagram, and a workflow step are not different *contents* in the same geometric space; they are objects in *different* spaces with incommensurable distance concepts. Two formulas may be close under operator-tree edit distance (syntactically similar) but far under dimensional-analysis distance (physically incompatible). Two authors may be close under citation-graph diffusion (frequently co-cited) but distant under concept-propagation distance (working in unrelated theoretical traditions). Conflating these into a single cosine similarity destroys the very information the distance is meant to convey.

Type-local metric families. For each concept type T we posit a *family* of admissible distance functions $\mathcal{D}_T = \{d_T^{(k)}\}_{k\in K_T}$, each reflecting a distinct projection of the concept's geometry:

- *Formula space* $(\mathcal{F}, \mathcal{D}_{\mathcal{F}})$: operator-tree edit distance (syntactic proximity); symbolic normalisation distance (do two forms reduce to the same normal form?); dimensional-analysis compatibility (do the physical dimensions align?); variable-role alignment (are the bound variables playing the same structural roles?); theorem-dependency overlap (how much of their proof graphs intersect?). Text-embedding distance is *not* in this family—it is a projection of the formula into the natural-language space, not a metric internal to formula space.

- *Bibliography space* $(\mathcal{B}, \mathcal{D}_{\mathcal{B}})$: citation-graph diffusion distance (random-walk proximity on the directed citation graph); concept-propagation distance (how many conceptual descendants does a work share with another?); co-author transport distance (optimal-transport over the author co-occurrence measure). Naive averaging of author vectors suffers from publication-count bias (a prolific author's centroid is pulled toward mass-produced work); diffusion-based metrics weight influence, placing Nash and Tate at high centrality despite sparse publication volume.

- *Procedural space* $(\mathcal{P}, \mathcal{D}_{\mathcal{P}})$: policy-graph edit distance (how many state transitions must be inserted, deleted, or substituted to transform one skill into another?); bisimulation distance (do the two skills agree on all observable branching outcomes up to depth n?).

Conditional distances. Because a concept participates in multiple spaces simultaneously—a formula has a syntactic face, a physical-dimensional face, a bibliographic face, and a proof-structural face—distances between concepts are *projection-conditional*:

$$d_T(a, b \mid \pi) \;=\; d_{\pi(T)}\big(\pi(a),\, \pi(b)\big), \tag{2.29}$$

where $\pi\colon T \to T'$ is a projection morphism and $d_{\pi(T)}$ is a metric in the target type's family. The same pair (a, b) of formulas may be ε-close under spoken-form projection (their verbal descriptions are nearly identical) yet far apart under symbolic-projection (their operator trees share no common subterm). This is not a contradiction; it reflects the genuine multi-faceted nature of mathematical identity.

Fiber-bundle and sheaf structure. The global CSP is therefore not a Cartesian product of type-spaces with one global metric, but a structure closer to a *fiber bundle*:

$$\mathcal{CSP} \;=\; \big\{\, (T_i, \mathcal{D}_{T_i}) \big\}_{i \in I} \quad \text{with projection morphisms} \quad \pi_{ij}\colon T_i \to T_j. \tag{2.30}$$

The base of the bundle is the type index set I; each fiber is a metric space $(T_i, \mathcal{D}_{T_i})$; and the transition maps are the projection morphisms. This is non-orthogonal: a formula's projection into spoken language produces text, but that text may back-project into the citation space (spoken names of authors), the theorem space (named theorems invoked), and the diagram space (described shapes)—axes that are coupled, not independent.

This is the precise sense in which classical embeddings fail: they presuppose a single vector space (a trivial bundle) and a global metric (a single section of the metric fiber), whereas the actual CSP is a non-trivial bundle with section-dependent metrics.

The consistency requirements across projections—a theorem's Lean proof term must be semantically equivalent to its LaTeX rendering, which must be entailed by its verbal description—are precisely the *gluing conditions* of a sheaf over the type-index category. Local consistency in each projection (each π_{ij} is individually well-formed) does not guarantee global consistency (all projections are mutually compatible), exactly as local sections of a sheaf need not glue to a global section. Checking global consistency is therefore a non-trivial algebraic operation, not a lookup.

The enriched category of concepts. Gathering these observations, the full CSP system is an *enriched category* **CSP**:

- *Objects*: concept types T_i (theorem, formula, citation, author, diagram, ...);
- *Hom-objects*: for each pair (T_i, T_j), a metric space $\mathrm{Hom}_{\mathbf{CSP}}(T_i, T_j) = (\mathcal{P}_{ij}, d_{ij})$ of projection morphisms with a cost metric measuring projection fidelity;
- *Enrichment base*: the category **Met** of metric spaces and non-expansive maps, so that composition of projections is a non-expansive map on hom-spaces—semantics cannot grow under composition.

In this category, distance between concepts is itself a *morphism-valued* quantity: asking "how far is formula a from formula b?" is not answered by a single real number but by a family of real numbers indexed by the projection through which you are measuring. Classical embeddings are the degenerate case of this enriched category in which all hom-objects collapse to $\mathbb{R}$ with the trivial metric and all projections compose to cosine similarity.

Star-shaped semantic regions over the fiber bundle. The fiber-bundle structure (2.30) reveals a further limitation of globally convex category regions. Within each fiber $(T_i, \mathcal{D}_{T_i})$, the type-local metric family defines its own betweenness and convexity; across fibers, the Manhattan inter-domain combination of (2.3) applies. By the correlation-collapse argument of section 4.5, globally convex regions in the fiber product are forced to be rectangular in fiber-index space—destroying the semantic correlations that make type-local geometry informative. A formula's syntactic proximity and its dimensional-analysis compatibility covary; a citation's venue prestige and its author's centrality covary; an engineering constraint's load-path category and its material compatibility covary. Axis-aligned rectangularity discards all such coupling.

Star-shaped semantic regions, centred at a *prototype concept* that maximally satisfies all relevant type-local similarity criteria simultaneously, are the geometrically faithful alternative. A concept c^* serves as star-centre for a semantic region R in the fiber bundle if every concept $c \in R$ is reachable from c^* via a monotone similarity path within the bundle—a multi-fiber generalisation of the star-shapedness condition of section 4.5. This ensures that: (i) prototype effects survive projection to each fiber, since $\pi_{ij}(c^*)$ is the star-centre of the projected region $\pi_{ij}(R)$; (ii) intersection of two star-shaped bundles sharing a prototype remains star-shaped, supporting compositional concept combination; (iii) correlated latent coordinates are represented faithfully, as the star-shaped region can be elongated along correlated fiber directions without forced axis-alignment. Adaptive retrieval geometries—context-induced weight deformations that reshape the bundle's effective metric—preserve this star-structure while reprioritising specific fibers, linking the theoretical framework directly to the constraint-preserving projections of section 2.8 and the transport-based retrieval of eq. (6.17).

3

Quality Dimensions and Domains

3.1 What is a Quality Dimension?

Quality dimensions are the basic axes along which an agent can discriminate stimuli. They are *cognitively primitive*: distinguishing between two points on a single dimension requires no further decomposition. Examples include:

- **Perceptual**: hue, brightness, saturation, pitch, loudness, temperature, hardness, weight.
- **Spatial**: the three Euclidean dimensions of physical location or shape parameters.
- **Temporal**: duration, onset, interval.
- **Abstract**: valence (pleasantness), dominance, arousal in the dimensional theory of emotion.

3.2 Integral versus Separable Dimensions

Following Garner's influential distinctions [Garner, 1974], dimensions are classified as:

Integral Dimensions that cannot be attended to independently. Hue and brightness are integral: an object that has no brightness has no colour at all. Distances along integral dimensions combine according to a Euclidean metric.

Separable Dimensions that can be attended to independently and are processed in parallel. Size and shape are largely separable. Distances along separable dimensions combine according to a city-block (Manhattan) metric.

This distinction has direct consequences for the shape of similarity gradients and is empirically testable via the *filtering paradigm* and *condensation task* from psychophysics.

3.3 Domains

Definition 3.1 (Quality Domain)**.** A *quality domain* is a set of dimensions that are *integral* with respect to one another. Formally, a domain

D_k is a metric space (D_k, d_k) that cannot be further decomposed into independent subspaces.

Representative domains and their constituent dimensions are listed in section 3.3.

Domain	**Dimensions**	**Metric**
Colour	Hue, Saturation, Brightness	Euclidean
Pitch	Chroma, Height (register)	Euclidean
Emotion	Valence, Arousal, Dominance	Euclidean
Space	x, y, z	Euclidean
Taste	Sweet, Sour, Salty, Bitter, Umami	City-block
Shape	Various curvature/moment parameters	Euclidean

Table 3.1: Example domains with their constituent dimensions and appropriate metrics.

3.4 Topology of Domains

Individual domains may have non-trivial topology. The hue dimension, for instance, is *circular*: red lies between violet and orange, and the two ends of the spectrum connect. This is reflected in the fact that the natural metric on hue is an arc-length metric on a circle S^1 rather than a Euclidean distance on $\mathbb{R}$.

More generally, a domain's topology encodes non-negotiable constraints on what transitions between quality values are possible. Gärdenfors calls the identification of appropriate topological structures for each domain the *topology problem* [Gärdenfors, 2000].

3.5 Hierarchical and Non-Linear Dimensions

Not all quality dimensions admit a continuous linear order. Many cognitively and computationally important domains are *hierarchically structured*: their points are organised as nested subdivisions rather than positions on a real line.

Example 3.2 (Document structure as a quality dimension)**.** A document is navigated by a coordinate tuple such as (chapter, section, subsection, paragraph). This tuple resembles a floating-point number in that it imposes a total order and permits arbitrarily fine subdivision, yet each component carries independent *semantic meaning* (a chapter boundary is qualitatively different from a paragraph break). The resulting dimension is *lexicographically ordered* rather than metrically linear, and the appropriate distance function must reflect the level at which two positions first diverge.

More generally, a *hierarchical quality dimension* is a rooted tree T whose nodes are quality values; the distance between two nodes $u, v \in T$ is defined as

$$d_T(u, v) = \ell(\mathrm{lca}(u, v) \to u) + \ell(\mathrm{lca}(u, v) \to v), \tag{3.1}$$

where lca denotes the least common ancestor and ℓ is a branch-length function encoding the semantic distance between adjacent levels. Equation (3.1) defines a valid ultra-metric, and the resulting *tree metric* can be embedded isometrically into ℓ^1 [Deza and Deza, 2009].

When multiple hierarchical dimensions must be combined—for example, a thesaurus (animal taxonomy), a spatial hierarchy (continent/country/city), and a temporal hierarchy (year/month/day)—the structures may not align. The *Gromov-Wasserstein* (GW) distance provides a principled way to compare and align two such metric-measure spaces without requiring a shared embedding:

$$d_{\mathrm{GW}}(\mathcal{X}, \mathcal{Y}) = \inf_{\pi \in \Pi(\mu,\nu)} \iint \left| d_{\mathcal{X}}(x, x') - d_{\mathcal{Y}}(y, y') \right|^2 d\pi(x, y)\, d\pi(x', y'), \tag{3.2}$$

where π ranges over all couplings of the measures μ on $\mathcal{X}$ and ν on $\mathcal{Y}$ [Goldfeld and Kato, 2022]. GW distance has been applied to align knowledge graphs, biological taxonomies, and language model representations—making it a natural tool for integrating heterogeneous conceptual spaces that share structural but not pointwise similarity.

Remark 3.3. The hierarchical dimension picture is consistent with the *topology problem*: the topology of a tree metric is discrete rather than continuous, which is the correct model whenever concepts have enumerable, named subdivisions rather than a continuum of values.

3.6 Generative Discovery of Quality Axes

Rather than hand-designing quality dimensions, one can treat a *variational autoencoder* (VAE) [Kingma and Welling, 2014] as a machine for discovering candidate CSP coordinates. The key insight is that a VAE trained on a corpus of scientific documents with a small latent bottleneck is forced to compress the essential structure of each document into a handful of real-valued variables. When the training objective is well-chosen, those latent variables may correspond to *disentangled, interpretable* quality dimensions: layout family, theorem density, citation community, notation regime, and the like.

A VAE latent axis is a good CSP coordinate iff nearby documents share stable structural transformations — i.e. iff the axis is *semantically continuous*.

The generative continuity criterion. A latent coordinate z_i is a *valid* CSP axis if and only if perturbing it continuously produces semantically smooth transitions in the decoded output:

$$z \longmapsto z + \varepsilon\, \mathbf{e}_i \quad \Longrightarrow \quad \text{decoded document changes smoothly.} \tag{3.3}$$

If tiny perturbations produce chaotic jumps—column count suddenly changes, equation environments appear or disappear—then z_i is not semantically stable and should be discarded or reparametrised. This criterion is empirically testable and provides a principled filter for separating meaningful CSP axes from noise dimensions.

The criterion is approximate: large decoder output changes are a necessary but not sufficient condition for semantic discontinuity—a single word substitution can reverse meaning while leaving most tokens unchanged. Calibration against an independent semantic signal (human typicality ratings, downstream task per-

Decoder sensitivity metric. The semantic contribution of each latent dimension is quantified by:

$$\text{importance}(z_i) = \left\| \frac{\partial\, \text{decoder}(z)}{\partial z_i} \right\|, \tag{3.4}$$

measuring how strongly the decoded output varies per unit of z_i. High-importance dimensions are primary CSP axis candidates; low-importance dimensions correspond to noise or redundant information and can be pruned without loss. The decoder sensitivity metric operates in output space rather than semantic space; its validity as a proxy for semantic continuity is contingent on the decoder being sufficiently faithful to make syntactic and semantic change co-vary consistently.

Reconstruction geodesic. The Euclidean distance $\|z_a - z_b\|$ in latent space is typically misleading because the manifold of valid documents is curved. A more faithful inter-document distance is the *reconstruction geodesic*:

$$d_{\text{geo}}(\mathbf{u}, \mathbf{v}) = \inf_{\gamma\colon [0,1]\to\mathcal{Z}} \int_0^1 \left\| \frac{d}{dt}\, \text{decoder}(\gamma(t)) \right\| dt, \tag{3.5}$$

where γ ranges over smooth paths from z_a to z_b in the latent space $\mathcal{Z}$. This geodesic measures the *semantic cost* of transforming one document into another and is the manifold-level analogue of the Gromov-Wasserstein distance introduced in eq. (3.2).

Discrete latent axes via VQ-VAE. Continuous latent spaces are appropriate for smoothly varying dimensions such as layout geometry, but *discrete* latent axes are better suited for categorical CSP dimensions such as proof strategy, citation role, or document genre. A *Vector-Quantised VAE* (VQ-VAE) [van den Oord et al., 2017] replaces the continuous posterior with a *codebook*:

$$z_q = \arg\min_{e_j \in \mathcal{E}} \|\hat{z} - e_j\|, \tag{3.6}$$

where $\mathcal{E} = \{e_1, \ldots, e_K\}$ is a learned finite vocabulary of semantic atoms. Each codebook entry is a candidate discrete CSP coordinate, analogous to the *named subdivisions* of a hierarchical dimension.

Simplicial structure of citation networks. Standard graph models treat citations as pairwise edges. However, scientific influence is frequently *multi-way*: a theorem may depend *jointly* on papers A, B, and C as an irreducible triple, not three independent pairwise links. This joint dependency is a *2-simplex*, and the full citation structure of a corpus is a *simplicial complex* Δ rather than a plain graph. Latent embeddings trained on this simplicial structure respect multi-way citation dependencies and produce richer CSP coordinates for the bibliography domain than any pairwise model.

A simplex $[A, B, C]$ encodes the fact that the theorem cannot be understood without all three references simultaneously—not just any two of them.

Remark 3.4. The generative-discovery programme is not in conflict with Gärdenfors's hand-designed dimensions: it is complementary.

Hand-designed axes encode *prior knowledge* about the domain, while data-driven axes may reveal *latent structure* that domain experts would never invent manually. The two sources are most powerful in combination: use generative discovery to propose candidate axes, then validate them against interpretability criteria (the generative continuity test and decoder sensitivity) and against prior domain knowledge.

4
Convexity, Prototypes, and Natural Categories

4.1 The Convexity Thesis

The central thesis of the conceptual-spaces framework is that *natural concepts* correspond to convex regions in a conceptual space.

Definition 4.1 (Convex Region)**.** A subset $R \subseteq D$ of a conceptual space is *convex* if for every two points $p, q \in R$ and every point b that is between p and q (in the sense of chapter 2), we have $b \in R$.

In Euclidean space, this is the standard definition: a region is convex iff it contains the entire line segment between any two of its members.

Example 4.2. The colour category RED is convex in colour space: any hue/saturation/brightness triple that lies on a path between two prototypically red stimuli is itself a shade of red (or at least a borderline red). By contrast, the ad-hoc category THINGS FOUND IN A KITCHEN is not convex in any plausible geometric space.

The convexity thesis is an empirical claim about human categorisation, not a logical necessity. Evidence in its favour includes:

1. Focal colours across languages tend to form convex regions in CIE colour space (Berlin & Kay, 1969; confirmed computationally in Regier et al. 2007).
2. Prototype effects (faster and more confident categorisation near the centre of a region) follow naturally if regions are convex and the prototype is the centroid.
3. Category learning experiments show that humans find convex categories easier to learn than non-convex ones [Feldman, 2000].

Remark 4.3 (Scope of the convexity thesis). The thesis holds most cleanly for *single-domain* categories (colour regions in CIE space, pitch regions in auditory space) where the intra-domain metric is Euclidean. For *multi-domain* categories combining correlated dimensions under the Manhattan inter-domain metric of eq. (2.3), global convexity is too strong: it forces category regions to be axis-aligned hyperrectangles, erasing inter-domain correlations. Section 4.5 develops the appropriate replacement: star-shaped regions relative to the prototype, which preserve prototype effects and correlation structure while remaining weaker than full convexity. The results of this section therefore apply directly to single-domain categories and as a first-order approximation for multi-domain categories; the star-shaped generalisation is the correct treatment for the latter.

4.2 Prototypes

Definition 4.4 (Prototype)**.** The *prototype* of a convex category R is its centroid (centre of mass):

$$\mathbf{p}^* = \frac{1}{\mu(R)} \int_R \mathbf{x}\, d\mu(\mathbf{x}), \tag{4.1}$$

where μ is an appropriate measure on D (e.g. the uniform measure or a probability measure reflecting exposure frequency).

Rosch's prototype theory [Rosch and Lloyd, 1978] is thus recovered as a theorem: if categories are convex, then there exist maximally typical members (the centroid and points near it), and typicality grades off with distance from the prototype.

4.3 Voronoi Tessellations and Category Boundaries

Given a set of prototypes $\{p_1^*, \ldots, p_k^*\}$, the most natural assignment of each point to a category is:

$$\mathrm{cat}(x) = \arg\min_i \delta(x, p_i^*). \tag{4.2}$$

This produces a *Voronoi tessellation* of D into cells $V_i = \{x \in D : \delta(x, p_i^*) \leq \delta(x, p_j^*) \text{ for all } j\}$. Each Voronoi cell is convex (in Euclidean space), which is consistent with the convexity thesis.

Remark 4.5. The Voronoi model predicts sharp category boundaries. In reality, human categorisation shows a *graded boundary region.* Extensions of the framework replace hard Voronoi membership with fuzzy membership functions or probabilistic assignment [Gärdenfors, 2014].

4.4 Conceptual Combination

Convexity also constrains conceptual combination. If RED and SQUARE are convex in their respective domains (colour and shape), their combination RED SQUARE is formed by taking the Cartesian product of the two regions in the joint colour-shape space. This product is again convex. More complex combinations (intersective adjectives, privative adjectives, metaphors) require geometric operations beyond simple intersection but remain tractable within the framework.

4.5 Beyond Convexity: Star-Shapedness and Correlated Manifolds

THE CONVEXITY THESIS rests on a hidden assumption: that the global metric δ is Euclidean, so that the line segment between any two category members lies in a geometrically natural region. Under the canonical two-level metric of eq. (2.3)—Euclidean within domains, Manhattan across domains—this assumption breaks down. Bechberger and Kühnberger [2017] prove that globally convex regions under a Manhattan inter-domain combination are necessarily *axis-aligned hyperrectangles*: every point that lies component-wise between two members of such a region must also be a member, regardless of whether the intermediate point occupies any region of the empirical distribution.

Correlation collapse. The failure mode is precise. Consider a conceptual space combining an `age` domain and a `height` domain for human children. The empirical distribution exhibits strong positive covariation: children are either both young and short, or both older and taller; the regions of young-and-tall or old-and-short are near-empty. A globally convex region containing a prototypically young-short child and a prototypically older-taller child must, by convexity, also contain the young-and-tall point—since it lies componentwise between the two members under the inter-domain Manhattan combination [Bechberger and Kühnberger, 2017]. The category region is forced to span combinations that no actual category member occupies, distorting prototype effects and erasing the correlation structure that gives the category its geometric meaning.

This is not a pathology of one domain pair. Colour spaces (hue and saturation covary), emotion spaces (arousal and valence are correlated across cultures), material-property spaces (hardness and melting point covary for crystalline materials), and engineering constraint spaces (load-path category and material compatibility covary) all exhibit the same inter-domain correlation structure. Global convexity in the Manhattan metric systematically destroys it, collapsing every correlated manifold into an axis-aligned box.

Star-shaped regions. Bechberger and Kühnberger [2017] replace global convexity with *star-shapedness relative to the prototype*. A region $R \subseteq D$ is star-shaped with respect to $p^* \in R$ if for every $x \in R$ and every $\lambda \in [0, 1]$, the interpolated point $(1 - \lambda)p^* + \lambda x$ lies in R—every point is *visible* from the prototype along a straight path:

$$\forall x \in R,\ \forall \lambda \in [0, 1]: \quad (1 - \lambda)p^* + \lambda x \in R. \tag{4.3}$$

Star-shapedness is strictly weaker than convexity (every convex set satisfies (4.3), but not conversely) yet sufficient to recover the core geometric commitments of the conceptual-spaces framework:

Prototype primacy. The prototype p^* is a parameter of the region, not a derived centroid. Typicality gradients emerge geometrically as membership decay along rays from p^*: the prototype is the most typical instance by construction, and graded membership follows from proximity along these rays.

Correlation preservation. A star-shaped region can be elongated along correlated axes—the high-probability region of a correlated bivariate Gaussian is star-shaped but not axis-aligned—capturing the genuine covariance structure of a category without the hyperrectangular collapse induced by global convexity.

Closure under intersection. The intersection of two star-shaped regions sharing a prototype p^* is again star-shaped with respect to p^*, since the intersection of two sets satisfying (4.3) for the same p^* also satisfies (4.3). Conceptual combination therefore preserves prototype structure within each constituent domain.

Fuzzy membership and α-cuts. In the graded extension, the region is specified by a *membership function* $\mu_R \colon D \to [0,1]$ satisfying $\mu_R(p^*) = 1$ and monotone non-increase along every ray from p^*. The α-cut

$$R_\alpha = \big\{x \in D : \mu_R(x) \geq \alpha\big\} \tag{4.4}$$

is star-shaped for every $\alpha \in (0,1]$, converting the topological property into a graded semantic structure consistent with psychologically observed gradience. This fuzzy formulation connects directly to confidence-thresholded retrieval: the confidence > 0.8 filter of eq. (6.17) is an α-cut at $\alpha = 0.8$ in the membership function over the retrieved concept bundle.

Remark 4.6. Star-shapedness makes the prototype operationally primary in a sense that convexity does not. A convex region has a well-defined centroid (4.1), but the centroid is a derived quantity that can fall outside a non-convex extension of the region. A star-shaped region is defined *relative to* its prototype: p^* is a constitutive parameter, not a secondary statistic. This is closer to the phenomenology of prototype theory [Rosch and Lloyd, 1978], where prototypes are cognitively antecedent and category boundaries are derived from them.

Star-shapedness as substrate for semantic manifolds. The Bechberger formalisation positions star-shaped regions as an implementable geometric approximation—a first step beyond axis-aligned convexity. The typed sparse vector space of section 2.6 and the fiber-bundle CSP of section 2.15 extend this: star-shaped regions in the fiber bundle respect the type-local metric families while preserving prototype structure across all projections simultaneously. The retrieval algebra of section 6.25 and the CSP-to-CSP operators of section 6.30 operate over star-shaped concept regions, using the closure-under-intersection property to guarantee that every operator composition (semantic slicing, Bayesian merge, normalisation) returns a region with a well-defined prototype and a semantically coherent extension. This chain—Gärdenfors convexity → Bechberger star-shapedness → typed fiber-bundle semantic manifolds—constitutes the geometric progression from the original framework to the operational CSP architecture developed in this manuscript.

5
Similarity and Distance

5.1 Similarity as Inverse Distance

A key appeal of the geometric framework is a principled account of *similarity*. Following Shepard's universal law of generalisation [Shepard, 1987], similarity between two stimuli p and q decays exponentially with distance:

$$s(p,q) = e^{-c\,\delta(p,q)}, \qquad c > 0. \tag{5.1}$$

This function is derived from a rational analysis of the problem of generalisation under uncertainty, and its Gaussian-decay form ($r = 2$ in eq. (2.1)) or exponential-decay form ($r = 1$) corresponds to whether the dimensions are integral or separable.

5.2 Tversky's Feature Model and its Geometric Counterpart

Tversky's contrast model [Tversky, 1977] expresses similarity as:

$$S(a,b) = \theta\, f(A \cap B) - \alpha\, f(A \setminus B) - \beta\, f(B \setminus A), \tag{5.2}$$

where A and B are feature sets. This model captures *asymmetric similarity* (a country may be judged more similar to a prototype nation than vice versa) and accounts for violations of the triangle inequality.

The conceptual-spaces framework reconciles Tversky's model with metric geometry by interpreting feature salience as *dimensional weights*. Asymmetry arises when the weight w_i applied to a dimension depends on which object plays the role of the referent.

5.3 Context Dependence and Dimensional Salience

A well-known challenge for purely metric models is that similarity judgements depend on context. "Piano" and "guitar" are judged similar in the context of "instruments" but dissimilar in the context of "things at a concert" (where size and price matter).

In the conceptual-spaces account, this is modelled by varying the weight vector $(w_1, \ldots, w_n)$ in eq. (2.1). Context selects a *relevance weighting* that highlights some dimensions and suppresses others, yielding the appropriate similarity ordering without changing the underlying space.

5.4 Metric Axioms and Their Violations

Standard metric axioms are:

$$\delta(p,p) = 0 \qquad \text{(reflexivity)} \tag{5.3}$$

$$\delta(p,q) = \delta(q,p) \qquad \text{(symmetry)} \tag{5.4}$$

$$\delta(p,r) \leq \delta(p,q) + \delta(q,r) \qquad \text{(triangle inequality)} \tag{5.5}$$

Human similarity data regularly violate (5.4) and (5.5). The conceptual-spaces framework accommodates these violations through context-dependent weighting (violating (5.4)) and through the use of non-metric *divergences* as distance functions in extended versions of the model.

5.5 Information-Theoretic Distance

An alternative grounding for distance in conceptual spaces draws on information theory. The *conceptual similarity* between categories A and B can be defined in terms of the maximum information content of the most specific category subsuming both [Resnik, 1995]. This connects conceptual-space geometry to distributional semantics: a word's meaning is located at the centroid of the conceptual-space points associated with its contexts of use.

5.6 Structural Importance and Document-Aware Similarity

Token-based embeddings treat all text passages as interchangeable, ignoring the rich structural hierarchy of a scientific document. The conceptual-spaces framework motivates a *structurally aware* importance weighting that assigns different salience to content depending on its position in the document tree.

Hierarchical structural weight. Let d denote the *depth* of a passage in the section hierarchy (0 for the title, 1 for a section heading, 2 for a subsection, and so on). The *structural weight* decays exponentially with depth:

$$w_{\text{struct}}(d) = e^{-\lambda d}, \qquad \lambda > 0, \tag{5.6}$$

giving the title weight 1, section headings weight $e^{-\lambda}$, and body paragraphs small weights. This matches the intuition that section titles, the abstract, and the conclusion carry disproportionate semantic load.

Reference-based importance. A passage is more important if it uses terms from the document's *controlled vocabulary*—the index, glossary, symbol list, and TOC. Define:

$$w_{\text{ref}}(x) = \sum_i P(t_i \in \mathcal{V}) \cdot \text{freq}(t_i, x), \tag{5.7}$$

where $\mathcal{V}$ is the controlled vocabulary and $\text{freq}(t_i, x)$ is the frequency of term t_i in passage x. This creates a *semantic gravity field*: passages that mention many index terms score highly regardless of their depth.

Multiplicative importance function. The overall importance of a passage is a multiplicative combination of structural, referential, mathematical, and citation signals:

$$I(x) = w_{\mathrm{struct}}(\mathrm{depth}(x))\cdot\big(1+\alpha\, w_{\mathrm{ref}}(x)\big)\cdot\big(1+\beta\,\mathrm{eq\_score}(x)\big)\cdot\big(1+\gamma\,\mathrm{cit\_score}(x)\big), \tag{5.8}$$

where $\alpha, \beta, \gamma > 0$ are configurable salience parameters. The equation density eq_score(x) measures how theorem-heavy the passage is; cit_score(x) reflects how frequently the passage is cited elsewhere in the document.

The multiplicative form ensures that interactions matter: a highly cited equation in the abstract receives far higher importance than the same equation buried in an appendix.

Query as transformation, not vector. A standard retrieval query is a point in the same space as documents. A structurally aware query is instead a *transformation operator*:

$$q = \big(\mathrm{boost}\colon \{k_1, \ldots\},\ \mathrm{penalise}\colon \{k_2, \ldots\},\ \mathrm{vocab}\colon \mathcal{V}_q,\ \mathrm{depth}\colon [d_{\min}, d_{\max}]\big), \tag{5.9}$$

which acts on the importance function rather than simply measuring cosine distance. This generalises the Tversky feature weights of eq. (5.2) to the full structural hierarchy.

Multi-view similarity fusion. For typed sparse vectors (section 2.6), similarity is best computed as a *weighted sum across views*:

$$\mathrm{sim}_{\mathrm{fused}}(\mathbf{u}, \mathbf{v}) = w_1 \cos(\mathbf{u}_{\mathrm{text}}, \mathbf{v}_{\mathrm{text}}) + w_2 \cos(\mathbf{u}_{\mathrm{struct}}, \mathbf{v}_{\mathrm{struct}}) + w_3 \cos(\mathbf{u}_{\mathrm{math}}, \mathbf{v}_{\mathrm{math}}), \tag{5.10}$$

where the three views correspond to natural-language content, document-structural position, and mathematical/symbolic content respectively. The weights w_i can be learned via a self-improvement loop: initialise heuristically, run retrieval tasks, measure ranking stability under perturbation, and accept updated weights when stability improves.

Remark 5.1. The Jensen-Shannon divergence $d_{\mathrm{JS}}(p \,\|\, q)$ is an appropriate base metric for the *histogram view* of a document: each passage is represented by a probability distribution over index terms, and JS-divergence is a symmetric, bounded, information- theoretically motivated distance between distributions. It complements cosine similarity, which works well for dense embeddings but poorly for sparse histograms.

6 Applications

The conceptual-spaces framework has moved well beyond theoretical cognitive science and is now an active area in artificial intelligence, natural language processing, and robotics. This section surveys the most significant lines of application.

6.1 Colour Naming and Cross-Linguistic Universals

Colour is the paradigmatic domain for conceptual-spaces research. The three-dimensional CIE $L^*a^*b^*$ colour space provides a perceptually uniform metric in which Euclidean distances approximate human discrimination thresholds. Regier, Kay, and Khetarpal [Regier et al., 2007] showed that the colour-naming systems of 110 languages can be predicted near-optimally by Voronoi tessellations of this space, providing strong evidence for the convexity thesis at a cross-linguistic level.

6.2 Grounded Word Meaning and Distributional Semantics

Distributional semantic models represent word meaning as vectors in a high-dimensional space derived from co-occurrence statistics. These vectors can be interpreted as points in a conceptual space, enabling the integration of linguistic and perceptual representations. Lenci [2018] reviews the connections between distributional semantics and prototype theory, arguing that the geometric structure of word-embedding spaces reflects the same convexity constraints predicted by the CSP framework.

6.3 Conceptual Spaces in Robotics and Human-Robot Interaction

Robots that must communicate with humans about objects and events in a shared environment need to ground linguistic categories in sensorimotor experience. Chella et al. [2008] implements a conceptual-space architecture in which a robot maps perceptual features (colour, shape, size) to quality dimensions and learns concept boundaries incrementally. The convexity constraint regularises learning: the robot generalises correctly to unseen instances because its categories are convex regions, not arbitrary decision boundaries.

6.4 Ontology and Knowledge Representation

Formal ontologies (e.g. OWL) represent concepts as classes in a description logic. Conceptual spaces provide a complementary, subsymbolic layer: concept membership is determined by proximity rather than by satisfying a set of necessary and sufficient conditions. Aditya et al. [2019] combines OWL ontologies with conceptual-space regions to handle the *open-world assumption* more gracefully, allowing soft membership and graded inheritance.

6.5 Conceptual Change and Metaphor

Conceptual spaces offer a geometric account of *metaphor*: a metaphor maps the structure of a source domain onto a target domain, preserving relational geometry. For example, "argument is war" projects the spatial topology of military engagement (attack, defence, territory) onto conversational structure. Zaltman's conceptual metaphor theory can be formalised as a *structure-preserving map* (morphism) between two sub-spaces of a larger conceptual space [Gärdenfors, 2014].

6.6 Commonsense Reasoning and Large Language Models

Recent work explores whether the internal representations of large language models (LLMs) exhibit the geometric properties predicted by conceptual-spaces theory. Probing studies suggest that linear probes can recover prototype structure from transformer activations, and that convex hulls of class representations are predictive of generalisation accuracy [Patel and Pavlick, 2022]. This opens the possibility of using CSP geometry as a diagnostic and design tool for neural language models.

6.7 Mechanistic Interpretability and the CSP Middle Layer

A central goal of mechanistic interpretability is to understand *how* neural networks represent concepts. A striking empirical finding is that large language models pack many more features into their activations than they have dimensions—a phenomenon called *superposition* [Elhage et al., 2022]. Individual neurons or linear directions in activation space simultaneously encode multiple unrelated features (*polysemanticity*), making direct inspection of weight matrices uninformative.

Sparse autoencoders (SAEs) attempt to undo superposition by projecting model activations into a large over-complete dictionary of *monosemantic* features, each of which hopefully corresponds to a single interpretable concept [Bricken et al., 2023]. The CSP framework provides a natural interpretation of this decomposition:

- The SAE dictionary is a candidate basis set for a *typed sparse vector space* (section 2.6): each dictionary vector is a candidate $\mathbf{e}_k$, and each document or prompt is represented as a sparse linear combination.

- Polysemanticity is the symptom of an *ill-chosen basis*—dimensions that conflate distinct quality regions.
- The CSP principle of disentangled quality dimensions is precisely what SAE decomposition attempts to recover.

The dark matter problem. Theoretical extrapolations from superposition theory [Elhage et al., 2022] suggest that current SAE methods recover a small fraction—potentially below 1%—of the features needed to fully account for model behaviour [Templeton et al., 2024]; this estimate follows from extrapolating dictionary sizes required for near-complete coverage against the total number of monosemantic features inferred from superposition scaling arguments. The CSP framework predicts such a shortfall: a polysemantic model must use exponentially many superposed directions to approximate what a typed sparse representation encodes with explicit named dimensions, so SAE recovery is bounded by the available dictionary size. This shortfall may reflect both engineering limitations of current SAE training (suboptimal initialisation, insufficient regularisation) and a more fundamental mismatch between dense continuous representations and the discrete structured nature of conceptual knowledge; the CSP programme addresses the latter, not as an alternative to engineering improvements, but as a complementary architectural strategy.

The CSP programme proposes a complementary strategy: rather than recovering a dense model's implicit features post hoc, build representations that are *explicitly* typed and sparse from the outset, so that the "missing" features are never superposed in the first place.

6.8 Theorem Provers as Conceptual Space Generators

Formal proof assistants offer a source of high-quality, machine-verifiable CSP coordinates for the mathematical domain. In Lean 4 with the Mathlib library [de Moura and Ullrich, 2021, The mathlib Community, 2020], every mathematical object has a precisely specified *type*, and every proof step is a *term* whose type is the proposition proved. This gives a direct mapping onto the typed sparse vector space of section 2.6.

Type classes as quality dimensions. Mathlib's algebraic hierarchy is a chain of *type classes*—each class adds one or more axioms (associativity, identity, inverse, distributivity, ...) to those inherited from its parent. Reading this hierarchy as a conceptual space yields:

Type class	**Added axiom**	**Quality dimension**
`Semigroup`	Associativity	Binary closure
`Monoid`	Identity element	Neutral element
`Group`	Inverse elements	Invertibility
`Ring`	Distributivity	Two-operation structure
`Field`	Division	Multiplicative inverses

Each level of the hierarchy corresponds to a *conceptual region* in algebraic space: the region of all groups is a convex superset of the region of all fields. Instance inference in Lean—automatically supplying the most specific applicable structure—is formally equivalent to *nearest-prototype retrieval* in the CSP sense.

Proof steps as difference vectors. Every tactic application in Lean transforms the current proof state (a goal) into a new proof state (remaining subgoals). Interpreting these states as points in the space of mathematical assertions, a proof is a *directed path* from the initial goal to the empty goal set. Each tactic step is a *difference vector*:

$$\mathbf{v}_{\text{after}} - \mathbf{v}_{\text{before}} = \text{semantic effect of the tactic.} \tag{6.1}$$

The Mathlib library, containing tens of thousands of lemmas, constitutes a *dense conceptual graph*: lemma application is edge traversal, and the full proof of a theorem is a path through this graph from axioms to the target statement. The graph structure is exactly the *citation graph* of section 3.6, realised at the level of formal mathematics rather than natural-language papers.

Example 6.1 (Lean-to-CSP mapping)**.** Consider the Lean 4 theorem

```
theorem add_comm (a b :  ) : a + b = b + a
```

Its typed sparse CSP vector (2.7) is assembled as follows:

Dimension	**Value**
`type`	`theorem`
`domain`	`natural_numbers`
`depends_on`	`{Nat.add, Nat.rec}` (Mathlib lemma IDs)
`proof_strategy`	`induction`
`lean4_term`	`fun a b => Nat.add_comm a b`
`proven_by`	`mathlib`
`cas-verified`	`true` (FriCAS: $a + b = b + a$ holds in $\mathbb{N}$)

The distance between two theorems is computed per component: `depends_on` overlap via Jaccard similarity, `proof_strategy` via a discrete metric over strategy types (`induction`, `rewrite`, `simp`, …), and `domain` via the conceptual-space metric of chapter 3. Each tactic application produces a difference vector (6.1) in this space; a complete proof is a path from the initial goal vector to the zero vector (empty goal set).

Symbolic computation as a CAS verification layer. Lean 4 handles logical deduction, but many scientific claims are *algebraic*—integrals, limit evaluations, derivations marked as "it trivially follows that"—where symbolic computation rather than logical proof is the appropriate verification tool. We extend Layer C to include a *computational veracity sub-layer*: each equation object E is paired with a computer-algebra derivation graph $G_{\text{CAS}}(E)$ produced by a system such as FriCAS (available through SageMath). If an author writes "it follows that $F = \ldots$", the CAS attempts to synthesise the omitted intermediate steps. The resulting equations become new typed objects with `f:` links to E in ScholarWiki format (section 6.26), effectively *decompressing* the reasoning for downstream LLMs and making hidden derivation steps first-class citizens of the CSP graph. The `cas-verified` attribute in the Layer C schema records whether the CAS derivation succeeded, partially succeeded, or failed—distinguishing algebraic facts from conjectures at the object level.

Agentic formalisation pipelines. Several production-grade pipelines now bridge the gap between LaTeX source and verified Lean code. **RepoProver** [Facebook Research, 2024] is a multi-agent scaffold for large-scale formalisation of mathematics textbooks: it extracts definitions and theorem statements from LaTeX, generates Lean skeletons, and iterates with a proof-search agent until the Lean compiler accepts the result. **Formath** [Formath Team, 2024] provides a staged TeX-to-Lean pipeline with source tracking and minimal human intervention. **LeanDojo** [Yang et al., 2023] adds an end-to-end environment for AI-assisted proof search with premise retrieval from Mathlib and lifelong learning across proof attempts. **Lean Copilot** [Yang et al., 2023] integrates LLM inference directly into the Lean IDE for tactic suggestion and goal completion. **TheoremLlama** [Sun et al., 2024] provides a Lean 4-specialised LLM trained via natural-language-to-formal bootstrapping and curriculum learning.

In the CSP framework, these tools map naturally onto existing operators: RepoProver and Formath populate the `theorem`-typed nodes of the CSP graph from LaTeX AST inputs; LeanDojo's premise retrieval is a typed graph traversal in eq. (6.16); Lean Copilot's tactic suggestions are difference vectors (6.1) proposed by a retrieval agent. Tools such as **Pantograph** (proof-state backtracking) and **APOLLO/HERMES** (compiler-error-guided proof repair) operate at the Layer C infrastructure level, enabling Monte Carlo search over the space of conceptual displacements guided by the global energy function E_{total} (eq. (6.27)).

6.9 Computing Systems as Conceptual Spaces

Human-computer interaction, operating systems, and document formats each constitute a conceptual space with characteristic quality dimensions and evolutionary trajectories. The following analysis, inspired by Gärdenfors's programme applied to software systems, identifies three mutually interacting spaces.

User-Computer Interaction Space. Dimensions characterising how users express and receive intent include:

Dimension	**Low**	**High**
Abstraction	Machine code	Declarative
Expressivity	Fixed cmds	Natural lang.
Feedback latency	Batch	Real-time
Control grain	Coarse	Fine (API)

Table 6.1: Interaction space dimensions.

PostScript occupies a distinctive region: high abstraction (device-independent graphics), high expressivity (Turing-complete stack operators), and medium control grain. Its innovation was to treat *page description as a programmable space*, making vector graphics first-class conceptual objects.

Operating System Space. The Unix philosophy clusters in the high-composability, high-openness quadrant; the *Embrace, Extend, Extinguish* (EEE) pattern describes a trajectory that begins at a high-openness point and moves toward lock-in—a directed path in this space.

Dimension	Low	High
Virtualization	Bare metal	Container
Composability	Monolithic	Microkernel
Access control	All-or-nothing	RBAC
Ecosystem openness	Proprietary	POSIX

Table 6.2: OS space dimensions.

Document / Format Space. PostScript and PDF illustrate opposite ends of the expressivity axis within this space: PostScript's Turing-complete expressivity yields high semantic density but unpredictable output; PDF's bounded interactivity sacrifices expressivity for reproducibility.

Dimension	Low	High
Semantic density	Binary/opaque	Annotated
Extensibility	Fixed fields	Open schema
Toolchain coupling	Portable	Locked
Standardisation	Proprietary	ISO-certified

Table 6.3: Document space dimensions.

The three spaces are not independent: they interact through *constraint propagation* (user intent propagates as constraints into the document space via OS mediation) and *semantic projection* (document features are projected onto OS resource attributes, e.g. via extended file attributes). A concrete example is the screw-replacement workflow: user intent ("replace all high-strength screws of type X") is expressed in the Interaction Space, mediated by OS file-search and pipe operators, and resolved in the Document Space by updating CAD and BOM files that encode screw properties as quality-dimension values.

6.10 Fractal Constraint Stacks

PostScript's dictionary stack offers a prototype for *context-aware search*: the active dictionary determines which names resolve to which procedures, and dictionaries are pushed and popped to create scoped name spaces. Generalising this to conceptual spaces yields the *Fractal Constraint Stack* (FCS) architecture[1]—a data structure in which each layer is a two-dimensional *constraint surface* over an embedding space.

[1] The term "fractal" here refers to the hierarchical, self-similar branching of the tree index (l, b, o), not to mathematical fractal geometry in the strict sense of scale-invariant self-similarity.

Constraint Surface. Replace the 1-D dictionary (key $\mapsto$ value) by a constraint matrix

$$M = \left[\mathbf{e}_i \,\|\, c_{ij}\right]_{i,j}, \tag{6.2}$$

where row i is a context slice (e.g. "screw properties in Chapter 3"), columns correspond to embedding dimensions (semantic, spatial, temporal), and each cell c_{ij} is a constraint function (e.g. tensile_strength $\geq$ 1200 MPa). This surface is the *conceptual-space analogue* of a Mixture-of-Experts gating layer: constraints act as gates selecting which regions of the embedding space are relevant.

Stack Operations. The stack supports three typed operations:

PUSH Freeze the current surface and open a new empty layer that inherits the embedding coordinates of its parent. *Semantics*: entering a document subsection, a product context, or a user task.

QUERY Project constraints downward through the stack, intersecting each layer's local constraints with the incoming query vector. *Semantics*: "find all screws in *this chapter* meeting strength criteria."

POP Collapse resolved constraints upward, summarising the layer's outcomes in the parent surface. *Semantics*: exiting a subsection, propagating updated part lists.

Fractal Position Encoding. Each element in the stack carries *dual embeddings*: a content embedding $\mathbf{e}_{\text{content}} \in \mathbb{R}^d$ (e.g. a BERT vector) and a structural embedding derived from its fractal coordinates (l, b, o)—level (depth from root), branch (index among siblings at level l), and ordinal (sequential position within the current level):

$$\mathbf{e} \;=\; \mathbf{e}_{\text{content}} + \gamma\big[\log(l+1),\; b/b_{\max}(l),\; o/n_l\big], \tag{6.3}$$

where $b_{\max}(l)$ is the maximum branch index at level l, n_l is the total number of siblings at level l, and γ projects the normalised structural triple into the same magnitude range as the content embedding. Each component of the structural triple is thus dimensionless and confined to $[0, 1]$: the $\log(l + 1)$ factor compresses deeper levels logarithmically (deep levels carry less absolute semantic significance than their distance from the root would suggest linearly), while $b/b_{\max}(l)$ and o/n_l provide normalised branch and ordinal coordinates. The coordinates (l, b, o) instantiate the tree-metric structure of chapter 3: level corresponds to the root distance $\ell(x)$, branch to the path from root to node, and ordinal to the local ordering within a level.

The original transformer positional encoding [Vaswani et al., 2017] uses sinusoidal functions of a single position index; (6.3) replaces that single index with a three-dimensional tree-metric coordinate, normalised so all three components lie in $[0, 1]$ before scaling by γ.

Bidirectional Constraint Propagation. The FCS supports two query directions that mirror the two directions of the betweenness relation in chapter 2:

- *Bottom-up* (refinement): "find all screws meeting specification in Chapter 3" — constraints narrow the search region as one descends the stack.
- *Top-down* (summarisation): "which chapters contain screws needing upgrade?" — outcomes propagate upward via POP, aggregating results across layers.

This bidirectionality is the computational realisation of the *morphism composition* described in the Personal Context section above: a user intent is a morphism from the global conceptual space to a local one, and the POP operation composes the inverse morphism back to the global level.

Implementation. The CSPChat / LATW pipeline [Kolbe, 2025] provides a concrete implementation of this architecture in TypeScript. Twelve processing modules run in priority order over a LaTeX source document; each module corresponds to one level of the constraint stack (document header, section, formula, table, theorem, paragraph). The *VQ-VAE* module at priority 4 assigns discrete codebook entries to semantic units before any structural token is consumed—realising the fractal position encoding of equation (6.3) as a learned quantisation over the three-axis space (object_type, proof_strategy, dependency_depth).

6.11 Document Cognition as Epistemic Compilation

The classical analogy between parsing and compilation provides a productive framework for understanding how raw documents are transformed into conceptual representations. Just as a compiler translates source code through successive *intermediate representations* (abstract syntax tree, static single-assignment form, machine-independent IR), a document processor transforms raw text through a cascade of increasingly semantic representations—from glyph sequences to CSP coordinates.

Compiler-CSP correspondence. The mapping between compilation stages and document cognition stages is surprisingly precise:

Compiler stage	**Document cognition analogue**
Lexer / scanner	Formula detector, citation scanner
Parser (LTAG / XTag)	Section/theorem/proof structure analyser
Type checker	Semantic linker (symbol scope resolution)
IR generation	CSP coordinate assignment
Optimisation	Dimension reduction, redundancy elimination
Code generation	TiddlyWiki tiddler / embedding output

Early-pass and late-pass processing. The two-pass architecture natural to compilers maps directly onto the document pipeline. An *early pass* handles syntax and structure recognition (formula detection, citation scanning, section boundary detection), while a *late pass* handles semantics and resolution (theorem dependency linking, bibliography look-up, symbol scope assignment). This separation mirrors the compiler distinction between a *front end* (language-specific, AST-level) and a *back end* (language-independent, IR-level). The parser stage in particular benefits from a formal grammar: *Lexicalised Tree-Adjoining Grammar* (LTAG) provides a mildly context-sensitive formalism that handles prepositional phrase attachment and other structural ambiguities that confound simpler chunkers. The XTag project demonstrated that LTAG parsers can be grounded in rich lexicalist feature structures—a precedent for feature-based lexicalised parsing of LaTeX-embedded natural language that maps naturally onto the typed basis vectors of section 2.6.

Two-pass processing resolves forward references: a formula can be recognised before its defining theorem is encountered.

Knowledge graph and path algebra. The compilation view aligns naturally with a *knowledge graph* (KG) representation. Each document element becomes a vertex; each semantic relation (cites, defines, uses, derives) becomes a typed edge. The resulting directed graph is a *quiver*, and the associated *path algebra* $k\mathcal{Q}$—whose basis is the set of all finite directed paths—gives an algebraic semantics to chains of reasoning. A proof of concept is provided by the Heim mass formula, a speculative physics theory with 30+ interlocking equations, 12 multiplet families, and a branching zone-extraction algorithm: once encoded as a quiver, each zone-extraction step becomes a morphism in the path algebra, validation test cases become queries over the graph, and the three-branch decision at step 8 becomes three distinct path families. A graph-based agent can pre-enumerate all paths, apply them in parallel, and compute error vectors between predicted and experimental masses—a level of complexity inaccessible to human manual tracking [Kolbe, 2025].

Attribute grammars as CSP extraction rules. An *attribute grammar* attaches synthesised and inherited attributes to grammar productions, enabling semantic information to flow both bottom-up (from leaves to root) and top-down (from root to leaves) [Gärdenfors, 2000]. This is the right formalism for CSP coordinate extraction rules: each syntactic construct (equation environment, theorem statement, citation command) carries both a *synthesised* CSP attribute (computed from its children) and an *inherited* context attribute (propagated from the enclosing section or document). The result is an *attributed document tree* whose leaf-to-root traversal fills in the typed sparse vector of eq. (2.4).

Side effects as latent CSP observations. Compiler optimisation treats observable effects (memory writes, I/O) as constraints on reordering transformations. Analogously, the "side effects" of reading a scientific passage—the concepts that must be activated to parse it, the inferences that must fire silently—are latent CSP observations. Capturing these latent activations (e.g. via probing classifiers applied to a reader model's intermediate states) reveals quality dimensions that are *used* even when not *mentioned*, providing a richer picture of the document's conceptual content than surface annotation alone.

6.12 Prompt Algebra and Semantic Interrogation

The constrained projection operators of section 2.8 form a genuine *algebra* whose composition rules determine how semantic information can be extracted, combined, and validated across views of the same document. This section develops that algebra and connects it to the compiler-CSP correspondence of section 6.11.

CSP as compiler frontend. Each compilation stage of section 6.11 corresponds to exactly one constrained projector P_i from eq. (2.11):

Compilation stage	**Constrained projector**
Symbolic preprocessing	Stable-anchor assignment (eq. (2.12))
Projection-specific parsing	P_{struct}: section/theorem/proof tree
Graph normalisation	P_{cite}: citation quiver extraction
Ambiguity preservation	Superposition over competing S_i
Semantic lowering	P_C: Layer C type-theoretic grounding
Latent-space encoding	CSP coordinate assignment (eq. (2.4))

The stages are *sequential* in the compiler metaphor but *parallel* in practice: each P_i operates independently on the same source document, and the final CSP coordinate is their join in the enriched category **CSP** of section 2.15.

Prompts as first-class semantic operators. A prompt is not a query string but a *semantic operator* with measurable geometric properties:

Orthogonality $\langle P_i, P_j \rangle = 0$ iff the two projectors activate disjoint latent dimensions; orthogonal prompts yield statistically independent information and are the right basis for parallel interrogation.

Projection overlap $\|P_i - P_j\|$ measures shared semantic content; high overlap flags redundant interrogation steps that can be merged.

Semantic entropy $H(P_i) = -\sum_k p_k \log p_k$ over the distribution induced on S_i; a low-entropy prompt produces a sharply peaked, high-confidence projection.

Ablative prompts (section 2.8) minimise $H(P_i)$ by design: constraining the model to a single register reduces uncertainty in the output distribution.

Latent coverage $\mu(P_i) = \mathrm{vol}(\mathrm{Im}\, P_i)/\mathrm{vol}(D)$; the fraction of document space faithfully reached by projector P_i.

Compositionality Whether $P_j \circ P_i = P_{j\circ i}$ for some composed operator $P_{j\circ i}$; composable projectors support modular pipeline construction.

Algebra of semantic interrogation. Let $\mathfrak{P} = \{P_i\}$ denote the set of all constrained projectors. Sequential composition:

$$(P_j \circ P_i)(d) = P_j\big(P_i(d)\big), \qquad d \in D, \tag{6.4}$$

equips $\mathfrak{P}$ with a monoid structure (identity = unconstrained prompt; composition = sequential constraint application). The *meet* $P_i \wedge P_j$ (simultaneous constraints, both must hold) and the *join* $P_i \vee P_j$ (either constraint, union of subspaces) extend this to a lattice over semantic perspectives.

This is a genuine *algebra of semantic interrogation*—not embedding-centric retrieval, but a structured calculus over the space of semantic viewpoints. The typed retrieval expression of eq. (6.16) is the special case where each P_i is a typed filter:

$$\texttt{retrieve}(\mathit{theorem},\ \mathit{related\text{-}to}(\mathit{formula}:X),\ \mathit{projected\text{-}as}(\mathit{lean})) = P_{\mathrm{lean}} \circ P_{\mathrm{formula}} \circ P_{\mathrm{theorem}}, \tag{6.5}$$

confirming that typed retrieval is a special case of prompt composition over the CSP knowledge graph. The prompt algebra therefore subsumes and generalises retrieval-augmented generation: RAG is the degenerate case $|\mathfrak{P}| = 1$ with no compositional structure.

6.13 Structured Document Processing

Business and scientific documents share a rich relational structure: an invoice references a delivery note, which references a purchase order, which references a contract. Each document type constitutes a *conceptual region* characterised by its indexing dimensions (contract number, order number, line-item codes) and its obligatory quality dimensions (amounts, dates, party identifiers). Conceptual spaces provide a natural representation for this structure, treating each document as a stimulus point in a product space of domain-level metrics [Gärdenfors, 2000].

Automatic document processing—using OCR tools such as MathPix or Tesseract to ingest scanned images—produces unstructured text that must be mapped back onto this structured space. The CSP approach proceeds in three steps:

1. **Dimension extraction.** A parser identifies quality values for each obligatory dimension (amounts, dates, reference codes) and locates the document's position in the corresponding hierarchical dimensions (chapter/section/paragraph for structured documents such as contracts or technical specifications).

2. **Relational linking.** Documents are connected by shared reference codes. These links correspond to *cross-space morphisms*: a purchase-order space and an invoice space share a dimension for the order number, and their intersection is non-empty precisely when a valid procurement relationship holds.

3. **Constraint verification.** Business rules (e.g. "an invoice's total must equal the sum of its line items") define *convex regions* of admissible document configurations. A document set that falls outside these regions is flagged rather than rejected outright, preserving the *soft membership* characteristic of conceptual spaces: a legal dispute may arise precisely from documents that are near-but-not-inside the admissible region.

The hierarchical coordinates introduced in chapter 3 are essential here: section numbers, exhibit labels, and line-item identifiers are discrete ordered dimensions whose tree metric (eq. (3.1)) determines which parts of two documents are *conceptually adjacent*.

6.14 Personal Context and Conceptual Space Merging

A conceptual space is not universal: the same stimulus may receive different coordinates in different agents' spaces. Personalisation—the mapping from a shared public space to a private one—is naturally modelled as a *conceptual space morphism*.

Consider a visual object-recognition system (e.g. YOLO) that labels an image region as "old gray-haired man". A user's personal conceptual space may contain a prototype $p^* = grandpa$ that is the nearest prototype to that label in a social-relation domain. The *personal overlay* is a function

$$\phi\colon \mathcal{C}_{\text{public}} \to \mathcal{C}_{\text{private}} \tag{6.6}$$

that maps public prototype labels to personally meaningful ones while preserving the relational geometry: objects that are similar in the public space remain similar in the private space.

This framework generalises naturally to *world models* [LeCun, 2022]: as an agent accumulates experience (through wearable cameras, spatial maps, interaction history), the private conceptual space is progressively refined. Objects acquire private identifiers ("my keys", "my bag") as the agent's spatial and ownership dimensions extend the public object taxonomy. Stacking private overlays on top of a shared perception module is equivalent to composing conceptual-space morphisms, and the composition is well-defined whenever the morphisms are structure-preserving maps between metric spaces.

6.15 Perceptual World Models as Active Conceptual Spaces

A KEY EXTENSION of the CSP framework replaces passive labelling with active hypothesis generation. Rather than assigning a fixed label to each perceived object, a *world model constructor* maintains a partial, uncertain scene graph and generates targeted retrieval questions to reduce uncertainty [LeCun, 2022].

Scene graphs as partial CSP instances. Each perceived entity is represented as a node carrying: a *confidence* value in $[0,1]$ for each asserted attribute; a *visibility source* tag (seen / inferred / missing); and a set of *query hooks*—questions whose answers would resolve the remaining uncertainty. Relations between entities are typed edges (spatial, functional, ownership, state). The resulting structure is a *partial CSP instance* in the sense of definition 2.1: a partial assignment of quality values, with unassigned dimensions playing the role of open constraints waiting to be propagated.

Soft type assignment. A single perceived object rarely belongs to exactly one concept. The CSP-compatible formulation replaces crisp type labels with a *soft type membership vector*:

$$\tau(x) = \left\{\mathsf{type}_i \mapsto p_i\right\}_i, \qquad \sum_i p_i \leq 1, \tag{6.7}$$

where p_i is the probability that entity x belongs to conceptual region type_i. A partially visible piece of furniture might carry $\tau = \{\text{desk} : 0.72,\ \text{counter} : 0.21\}$. This is the continuous analogue of prototype distance in standard CSP: proximity to prototype i determines p_i, and the distribution τ is revised as new observations arrive—exactly the update step of Bayesian constraint propagation.

Templates as structured curiosity generators. Commonsense knowledge—the expectation that a *workstation* contains a monitor, a keyboard, and a chair—can be encoded as a *template*: a partial subgraph specifying expected entities, expected relations, and question generators for missing slots. Templates are *not* fixed schemas; they are negotiable hypotheses that inject new questions into the retrieval loop when activated and are revised when evidence contradicts their predictions.

This is formally equivalent to *abductive constraint propagation*: selecting the most probable template means finding the partial assignment that best extends the current CSP instance to a consistent full assignment under the template's prior.

Active perception loop. The resulting pipeline is a closed epistemic loop:

perception $\rightarrow$ partial CSP $\rightarrow$ template matching $\rightarrow$ expectations + questions $\rightarrow$ retrieval $\rightarrow$ graph update $\rightarrow$ perc

Each cycle reduces the total constraint uncertainty of the scene graph, implementing a form of active learning over perceptual quality dimensions [Gärdenfors, 2014]. The vision module acts not as an annotator but as a *hypothesis generator*: its output is not a label but a set of candidate partial assignments together with the questions required to discriminate among them.

6.16 Mixture of Closures: Constraint-Coupled Expert Systems

The Mixture-of-Experts (MoE) architecture routes each input to a single expert network. A more powerful variant replaces routing with *constraint coupling*: each expert maintains its own constraint satisfaction loop and the experts share a common *intermediate representation* (IR). We call this a *Mixture of Closures* (MOC).

Expert as CSP closure. Each expert E_k in the MOC is a triple

$$E_k = (\mathcal{D}_k,\ \mathcal{R}_k,\ \mathsf{runCSP}_k), \tag{6.8}$$

where $\mathcal{D}_k$ is the expert's domain model (a typed sparse vector space, section 2.6), $\mathcal{R}_k$ is its constraint store, and runCSP_k is its convergence loop that iteratively propagates constraints until reaching a fixed point or detecting a violation.

Shared IR and constraint coupling. Experts operate on a shared IR—a mini-CSP instance whose type system is grounded in the global key universe Keys of eq. (2.4): every entity, property, and relation in the shared IR is a typed key from this universe, so cross-expert communication is structurally well-typed. When two experts operate with different local key subsets $K_1, K_2 \subseteq \mathsf{Keys}$, inter-expert translation uses the projection morphisms π_{ij} of eq. (2.29): Expert k's contribution is projected from its local type-space $(T_k, \mathcal{D}_{T_k})$ into the shared IR's type-space via the appropriate morphism. Unit and scale mismatches between experts are handled by the normalisation operator (C) of section 6.30: canonicalisation maps domain-local scales to a shared canonical form before any constraint is evaluated globally. Violations detected by one expert are posted to the shared IR as *patches*: incremental updates that other experts must reconcile with their own constraint stores. This coupling is formally a *hypergraph constraint*: the admissible assignments must satisfy all experts' constraints simultaneously, not merely each expert's local constraints in isolation.

Debate as stochastic constraint search. A multi-agent *debate* layer can be understood as stochastic search over the space of partial assignments to the shared IR. Each debating agent proposes an assignment (a "position"), and the CSP layer checks consistency. The two layers are complementary: without the debate layer, the CSP solver may get stuck in a local minimum; without the CSP layer, debate produces inconsistent or contradictory conclusions. The MOC architecture therefore generalises the Fractal Constraint Stack of chapter 6: the constraint surfaces of the FCS become expert domains, and the stack operations become the shared IR update protocol.

Cross-expert interference and global arbitration. The hardest case in MOC is *antagonistic coupling*: two expert constraint stores, each individually satisfiable, that are jointly infeasible. A canonical example from manufacturing engineering illustrates the pattern. An adhesion expert requires roughness$(A) > r_0$ (rough surface maximises wetting and surface energy); a mechanical expert requires roughness$(A) < r_1$ (smooth surface minimises stress concentration at the interface); with $r_1 < r_0$, the two constraints cannot be satisfied simultaneously. A thermal expert may add a third incompatible requirement (expansion freedom breaks electrical contacts). These are not *local* constraint violations detectable by any single expert's convergence loop; they emerge only when the shared IR is examined globally.

The MOC architecture handles this by elevating the conflict to a *global CSP* layer:

$$\text{find } S \text{ such that } \forall k\colon \mathsf{violations}(\mathcal{R}_k, S) \leq \varepsilon_k,$$

where $\varepsilon_k \geq 0$ allows soft constraints and explicit trade-offs. The debate layer (section 6.16) then becomes a stochastic explorer over the Pareto surface of this multi-expert optimisation: each debating proposal is a candidate trade-off point, and the CSP layer determines whether that point is feasible and how far it lies from each expert's constraint boundary. Crucially, each constraint violation maps unambiguously to its source: violation $\to$ constraint $\to$ expert $\to$ entity $\to$ cause, providing full auditability without post-hoc explanation. This is the MOC analogue of the *minimal FMEA*: failure modes are not enumerated by human analysts in advance but generated automatically as the constraint boundaries of the global CSP.

6.17 Coordinate-Adaptive Sparsification and Matryoshka Representations

A GENERAL DESIGN PRINCIPLE emerges from the analysis of adaptive sampling systems: *good conceptual spaces are those in which the sig-*

nal becomes compressible under simple sampling rules. Two concrete examples illuminate this principle from complementary directions.

Coordinate-adaptive sampling. A train data recorder stores state samples (t_i, x_i, v_i) with variable spacing: at high speed the sampling interval is proportional to distance covered; at low speed to time elapsed. The key transformation is the *arc-length reparameterisation*

$$T: t \;\mapsto\; s = \int_0^t v(\tau)\, d\tau, \tag{6.9}$$

under which the trajectory becomes compressible: a uniform grid in s captures the same geometric information as a denser non-uniform grid in t. The sampling density

$$\rho(t) \;\propto\; \frac{1}{v(t)} \tag{6.10}$$

is inversely proportional to speed—equivalently, proportional to local information density. Analogous systems appear across signal processing: event-based cameras emit spikes only when log-intensity changes by ε; adaptive ODE solvers use step sizes proportional to local curvature; GPS trajectory simplifiers (Douglas–Peucker) keep only vertices whose removal would introduce more than ε geometric error.

The unifying abstraction is: for a signal f and transformation T, find T such that $T(f)$ is maximally smooth, then sample uniformly in T-space. This is a *metric selection problem* in the CSP sense: the "right" quality dimensions are those in which the structure of interest becomes explicit and compressible.

Matryoshka Representation Learning. Matryoshka Representation Learning (MRL) [Kusupati et al., 2022] encodes the same principle in vector embeddings. An MRL-trained model packs the most essential information into the *earliest* dimensions of each embedding vector, with subsequent dimensions capturing progressively finer detail, so that any prefix $\mathbf{v}_{:k}$ remains a valid, high-quality embedding at resolution k. The multi-scale training loss enforces this simultaneously across a set K of granularities:

$$\mathcal{L}_{\text{MRL}} = \sum_{k \in K} w_k \cdot \mathcal{L}(\mathbf{v}_{:k}). \tag{6.11}$$

The correspondence with coordinate-adaptive sampling is precise (see section 6.17): both systems implement an *importance ordering* over representational atoms so that any prefix of the ordered sequence is a valid, coarser-grained representation of the whole. The train recorder is "Matryoshka without explicit indexing": the upgrade is to make the ordering explicit and to optimise for prefix quality at multiple scales simultaneously.

Implication for CSP. In the CSP framework, quality dimensions are not equally informative. An importance ordering on dimensions—derived from decoder sensitivity (eq. (3.4)), structural weight (eq. (5.6)),

Matryoshka embeddings	**Coordinate-adaptive sampling**
Early dimensions = coarse info	Sparse samples = coarse trajectory
Later dimensions = refinements	Dense samples = local corrections
Prefix $\mathbf{v}_{:k}$ = valid repr.	Prefix S_k = valid trajectory approx.
Multi-scale loss (6.11)	Multi-scale fidelity constraint
Dimension index = importance rank	Sample index = importance rank (implicit)

Table 6.4: Structural correspondence between Matryoshka embeddings and coordinate-adaptive sampling. Both implement hierarchical information packing over an ordered atom set.

or arc-length reparameterisation (eq. (6.9))—enables *truncatable conceptual representations*: a k-dimensional prefix of the ordered quality space is a valid, coarser-grained model of the full concept. This connects the generative discovery of quality axes (section 3.6) to the practical engineering question of how many dimensions to retain at inference time, and gives a theoretical grounding to the observation that the earliest principal components of a latent space carry disproportionately much conceptual information.

6.18 LLM-Driven Axis Induction and the Canonicalization Problem

A FUNDAMENTAL OBSERVATION about large language models is that they do not merely learn *coordinates* in a pre-specified quality space—they implicitly learn the *axes themselves*. This process of *axis induction* discovers which distinctions matter for a given domain by exposure to training data, and its formalisation connects directly to the generative programme of section 3.6.

Axis induction vs. coordinate assignment. Classical CSP theory takes quality dimensions as given and places concepts at coordinates within them. Axis induction inverts this: given a corpus, a model induces the dimensions $\{D_i\}$ that best explain the observed similarity structure—a latent semantic axis discovery. The distinction matters practically: when asked "what dimensions characterise the space of railway power supply designs?", an LLM can *propose* axes (thermal envelope, voltage range, certification level, form factor) rather than merely scoring a design on pre-specified axes. This makes the LLM a co-designer of quality space, not merely a feature extractor.

VQ-VAE as axis discretization. A vector-quantized variational autoencoder (VQ-VAE) [van den Oord et al., 2017] learns a discrete codebook $\{c_1, \dots, c_N\}$ of prototype vectors such that any input is approximated by its nearest codebook entry. In the CSP interpretation, each codebook entry is a candidate quality-axis direction, and the quantization step is a *prototype snap*—assigning each concept to its nearest quality prototype and thereby partitioning the continuous latent space into convex regions. The VQ-VAE commitment loss ensures the codebook covers the relevant regions, implementing an adaptive Voronoi tessellation of quality space.

The axis canonicalization problem. When axis induction is run independently on overlapping corpora—or by multiple LLM instances—the resulting axis sets may disagree. Two runs may discover functionally equivalent axes under different labels ("thermal envelope" vs. "heat budget"), rotated orientations, or merged/split configurations. The *axis canonicalization problem* asks: given two independently induced axis sets, find the structure-preserving alignment that identifies equivalent axes and detects genuinely novel ones. This is a special case of Gromov-Wasserstein alignment [Goldfeld and Kato, 2022]: find the map T minimising the distortion of pairwise inter-axis distances. A solved canonicalization yields a domain-level *canonical quality space* independent of the particular LLM run that generated it—convergent quality dimension discovery from multiple independent induction processes.

6.19 *Theorem Introduction as Manifold Surgery*

In the incremental learning model of conceptual spaces, new knowledge is typically modelled as the addition of a point to a fixed manifold, or a smooth displacement of an existing prototype. The introduction of a significant theorem is qualitatively different: it is a *topological event* that alters the global structure of the concept manifold.

Phase transitions in conceptual topology. Consider the space of all known mathematics before and after the Prime Number Theorem. The theorem does not merely add a new point; it establishes a bridge between prime distribution (number theory) and complex analysis, linking two previously distant regions. Such bridges can add handles (connecting separate topological components), collapse dimensions (revealing that two apparently independent axes lie on a lower-dimensional submanifold), or refine convex regions into non-convex parts separated by a new distinction. These are precisely the moves of *Morse theory*: handle attachments triggered by passing through a critical value of a smooth function on the manifold. Theorem introduction is the conceptual analogue of passing through a critical point—the topology changes discontinuously at the moment the theorem is accepted.

Hybrid dynamical systems view. Conceptual space evolution can be modelled as a *hybrid dynamical system*: a continuous flow (gradual prototype drift from incremental learning) punctuated by discrete events (new definitions, proved theorems, refuted conjectures). The continuous part smoothly updates prototype locations and region boundaries; the discrete part performs *manifold surgery*—adding or removing topological features. Reset maps in hybrid system theory capture this: at each discrete event the state space is reconstructed, not merely the state within a fixed space.

Citations as nonlocal forcing operators. A citation is not merely a bibliographic pointer; it is a *nonlocal forcing operator* on the author's conceptual space. By invoking a reference, the author imports an entire algebraic framework—its definitions, theorems, and implicit assumptions—into the local reasoning context. This import is (i) *nonlocal*, because the imported framework may have been developed in a completely different quality-dimension space; (ii) *selective*, because only the sub-framework relevant to the citing paper is imported, corresponding to a projection of the cited paper's CSP onto the local axes; and (iii) *topology-altering*, because the imported definitions may introduce new quality dimensions or collapse existing ones. Detecting which topological events are triggered by which citations—and whether the stated Layer B process tag (section 2.7) matches the actual topological change at Layer C—would constitute the *conjectured hidden novelty detector* described in section 2.7: a research programme that the manifold-surgery framework makes tractable, but whose realisation depends on a not-yet-specified inter-layer composition rule and remains to be empirically validated.

6.20 Document Object Ontology and the Two Semantic Graphs

A PRECISE DOCUMENT REPRESENTATION requires not just quality dimensions but a *typed object ontology*: a closed system of document entity types with well-specified identity, relations, and graph membership. The architectural key is that every document gives rise to two orthogonal graphs that must be kept strictly separate.

Structure graph and semantic graph. The *structure graph* encodes layout and reading order: sections contain paragraphs, paragraphs contain content nodes, and cross-references create directed edges between numbered objects (equations, figures, tables). The *semantic graph* encodes meaning and reference: a symbol is *defined* in one node and *used* in many; a citation *refers to* a bibliographic entry while *occurring in* a specific content node; a proof *proves* a lemma. Collapsing these two graphs creates most downstream ambiguities: a citation appearing in section 3 is structurally adjacent to the equations there, but semantically it refers to an external work that may operate in entirely different quality dimensions.

Symbols as internal basis vectors. Every document generates its own sub-language by introducing symbols, defining their domains and units, and constraining their relationships through equations. This sub-language is the document's *internal quality basis*: each defined symbol is a candidate basis vector, and each equation involving symbols

$s_1, \ldots, s_k$ is a constraint hyperedge relating those vectors. The bibliographic graph, by contrast, provides *external anchors*: each cited work is a bound vector in the global knowledge space, and the citation is a cross-space projection from the cited paper's quality space onto the citing document's local axes. This separates the internal basis (symbols $\to$ quality dimensions) from external anchors (citations $\to$ cross-space projections), realising the affine structure of eq. (2.6) at the document level.

Root object types. The minimal closed type system for a scientific document includes: `Document` (root; has authors, sections, indexes), `Author` (carries ORCID; anchor for bibliographic layer), `BibEntry` / `Citation` (definition vs. usage strictly separated), `Section` (level, title, number; hierarchically nested), `Symbol` (notation, domain, description; defined once, used many times), `MathExpression` / `Equation` (extends symbol usage with AST; equations carry dependency edges `dependsOn` and derivation edges `derivesFrom`), `Figure` / `Table` / `Listing` (numbered objects with caption and cross-reference edges), and `Lemma` / `Theorem` / `Proof` (formal logic objects; a proof carries a `proves` edge). This ontology is not format-derived—the same types apply whether the source is PDF, LaTeX, Markdown, or spoken text—but content-semantic, capturing *what* each object means rather than how it is encoded.

6.21 Multi-Agent Expert Systems as CSP Pipelines

The typed sparse CSP framework extends to *distributed expert systems*: organisational structures in which specialised agents—each with its own domain model— collaborate on problems that no single agent can solve alone. CSP provides both the communication protocol and the backbone for traceable collective reasoning.

Expert as CSP closure. A domain expert is not a set of neural weights but a closed-loop system:

$$\mathsf{Expert} = (\mathsf{State}, \mathsf{Skills}, \mathsf{Processes}, \mathsf{Memory}, \mathsf{Tools}), \tag{6.12}$$

where State is the evolving knowledge base (facts, models, constraints), Skills are versioned executable procedures with typed I/O, Processes are CSP modules that transform state with strict contracts, Memory is the structured archive of past processing episodes, and Tools are external computational resources. The LLM is the inference engine; the expert is the system built around it.

Structured inter-expert messages. Communication between experts must not be free-form natural language, which permits inconsistent commitments and untracked hallucinations. Instead, each message is a *typed CSP patch*: a structured update to the shared IR specifying source, target, message type, and a typed payload. The receiving expert's CSP module validates the patch against its constraint store, augments the shared IR, and returns a typed response. Every transformation is explicit, every decision is stored, and every disagreement surfaces as a constraint violation rather than a conversational ambiguity—exactly the traceability guaranteed by the MOC architecture of section 6.16.

Self-improvement as state evolution. Deep domain expertise accumulates from structured experience, not from scale alone. The self-improvement loop is:

$$\text{Experience} \rightarrow \text{Reflection} \rightarrow \text{Skill update} \rightarrow \text{State update} \rightarrow \text{Experience.}$$

A design failure is captured as a typed event, transformed by a reflection agent into a skill patch, and merged into the relevant skill file via a version-controlled diff. The constraint that triggered the failure is added to the constraint store. Over many episodes the constraint store converges toward the implicit knowledge of a human domain expert—but in an explicit, auditable, diffable form. A shared *mission state* encodes the organisation's overarching constraints (e.g. "onboard railway products only, excluding trackside infrastructure"), ensuring that all expert outputs remain globally consistent and every decision can be replayed from its typed inputs.

Three-layer memory architecture. The Memory component of eq. (6.12) is not a monolithic store. A well-functioning expert system requires three strictly separated memory layers, which current industry systems routinely conflate:

Semantic memory Typed concept objects with their relations—theorems, formulas, entities, constraints, documents. This is the declarative CSP (section 2.14): what things *are*. Structure: $\{\texttt{concept} : c, \texttt{relations} : [r_1, \ldots]\}$.

Execution memory The auditable trace of processing episodes—which tool was called, with what input, at what step, producing what output. This is the procedural CSP (section 2.14): what was *done*. Structure: $\{\texttt{agentStep} : n, \texttt{tool} : t, \texttt{input} : x, \texttt{output} : y\}$.

Epistemic provenance The confidence and origin of every semantic memory entry—which source stated it, how confidently, whether it has been human-verified, and which constraint violations it has survived. Structure: $\{\texttt{source} : s, \texttt{confidence} : p, \texttt{humanVerified} : \top/\bot\}$.

Conflating these three layers is the primary source of untraceable hallucinations in LLM-based agents: a model may report a fact (semantic layer) that was never grounded in a source (provenance layer) and arrived via an execution path that was never recorded (execution layer). The typed sparse vector framework of section 2.6 enforces the separation structurally: `theorem`-typed vectors carry semantic content, `process`-typed vectors carry execution tags, and `pointer`-typed vectors carry provenance links.

Specialised agent roles. The five-component expert tuple (6.12) instantiates as a set of specialised agents with distinct input/output contracts and CSP-layer responsibilities:

Agent	**Input / Output**	**CSP equivalent**
Structural Extractor	PDF/LaTeX glyphs → ScholarWiki records	Fractal Constraint Stack (section 6.10) + LTAG parser
CAS Critic	Equation objects → derivation graphs	Layer C CAS sub-layer; `cas-verified` attribute
Formal Proof Synthesizer	Theorem stubs → verified Lean terms	RepoProver / Formath / LeanDojo pipeline (see above)
Contextual Cartographer	Citation edges → typed modalities	`CIT_AXIOM/METHOD/CONTRAST/RECONTEXT` (eq. (2.9))
Master Orchestrator	Partial CSP views → fused posterior	CSP-to-CSP operator composition (section 6.30)

Table 6.5: Specialised agent roles in a CSP-based document understanding system and their mapping to existing CSP constructs.

The orchestrator's composition policy can be trained via reinforcement: the D_{CSP} distance metric of eq. (6.25) (see section 6.30) provides a deducibility reward signal—the reduction in D_{CSP} after each operator pass—analogous to the GRPO reward used in self-evolving graph memory architectures.

6.22 *Projectional Knowledge Systems: FMEA as a CSP Query Basis*

A complex engineered artifact cannot be understood from a single representation. A mechanical bracket simultaneously has a geometry, an assembly sequence, a set of failure modes, a tolerance chain, a set of required manufacturing steps, and a list of maintenance actions—each representation useful for a different class of questions, each a projection of the same underlying conceptual object. This *projectional* view of knowledge is precisely the structure of a CSP with multiple measurement bases (cf. section 2.13).

FMEA as structured projection. Failure Mode and Effects Analysis (FMEA) is standardly presented as a risk table. In the CSP framework it is better understood as a *structured projection system*: a mapping from an artifact's conceptual space onto a failure-mode subspace whose axes are

FMEA dimension	**CSP interpretation**
Failure mode	Constraint violation
Cause	Missing or violated dependency edge
Effect	Propagation in the causal graph
Detection	Observability of the violation
Mitigation	Constraint-satisfying transformation

Each FMEA row is a constraint violation together with its causal ancestry and its consequences in the artifact's constraint graph. What the table format hides is that these rows are not independent: a single causal event can propagate through the graph and appear as multiple failure modes in different projections. A CSP representation makes this explicit by storing the violation once (at the constraint boundary) and deriving the FMEA rows by graph traversal.

The assembly-accessibility problem. A paradigmatic class of failures invisible to pure geometry analysis are *assembly-accessibility constraints*: situations where a fastener or component cannot be reached by any physically realisable tool in any physically realisable assembly sequence—even though the part is geometrically valid. Formalised as a CSP, the query is: does there exist a feasible tool path satisfying

$$\exists\,\text{path}(T) \text{ from tool approach to fastener location such that collisionFree(path)} \wedge \text{clearance(path)} \geq c_0?$$

If no such path exists, the corresponding FMEA entry is generated automatically: *failure mode* = non-assemblable, *cause* = inaccessible fastener, *effect* = production halt. The constraint is not derivable from the geometry graph alone; it requires the *assembly-process projection* of the artifact's conceptual space, making it a genuine completion task for a world-model architecture [LeCun, 2022].

Multi-projection queries and LeCun's world model. The FMEA-as-projection architecture generalises immediately: any question about an artifact is a query on a specific projection of its conceptual space. "Which materials can be substituted without violating any constraint?" queries the materials-and-tolerances projection. "In what order must sub-assemblies be inserted?" queries the process-sequence projection under collision constraints. LLMs can traverse between projections—geometry → assembly → failure → mitigation—far more reliably than humans can, because the projection graph is explicit and the constraint propagation is checked automatically. The world model that Gärdenfors's framework calls for is not a neural network that interpolates between training examples; it is a *multi-projection CSP* that supports inference, counterfactuals ("what if the hole moves 3 mm?"), and constraint repair—with FMEA as one of its principal query bases.

6.23 Holographic Memory and Attention as CSP Retrieval

THE MECHANISTIC INTERPRETABILITY PROGRAMME (section 6.7) asks which features and circuits inside a neural network correspond to identifiable concepts. A complementary question asks about the *storage mechanism*: how are concept representations encoded in weights, and what retrieval operation recovers them? The answer connects CSP to classical associative memory and holographic storage.

Hopfield networks as holographic superposition. A modern Hopfield network *stores* patterns $\{\mathbf{x}_i\}$ in a weight matrix built by superposition:

$$W = \sum_i \mathbf{x}_i \mathbf{x}_i^\top. \tag{6.13}$$

This is *hologram-like*: each pattern contributes a rank-1 interference term, and the total W is their superposition. *Retrieval* given a query $\mathbf{x}_q$ is the matrix-vector product followed by normalisation:

$$\hat{\mathbf{x}} = \mathrm{softmax}(\beta\, W \mathbf{x}_q) = \mathrm{softmax}\left(\beta \sum_i (\mathbf{x}_q \cdot \mathbf{x}_i)\, \mathbf{x}_i\right), \tag{6.14}$$

which is a softmax-weighted sum of the stored patterns—the CSP operation of finding the nearest prototype, softened to a convex combination. Ramsauer et al. [2021] proved that the fixed-point update rule of the continuous-state Hopfield network is exactly equivalent to the key-value attention mechanism of Vaswani et al. [2017]. The connection becomes explicit when keys $\mathbf{k}_i$ and values $\mathbf{v}_i$ are identified with stored patterns $\mathbf{x}_i$:

$$\hat{\mathbf{v}} = \sum_i \mathrm{softmax}_i\left(\frac{\mathbf{x}_q \cdot \mathbf{x}_i}{\sqrt{d}}\right) \mathbf{x}_i, \tag{6.15}$$

which is, up to the $1/\sqrt{d}$ temperature scaling, the scaled dot-product attention output [Vaswani et al., 2017, Ramsauer et al., 2021].

Where the hologram analogy breaks. The optical hologram stores views of a *single* physical scene from multiple angles; coherence across views is guaranteed by physics. In a language model, inputs are independent sequences with no shared physical referent, so the "hologram" analogy fails at the point of superposition: $\sum_i W(\mathbf{x}_i)$ becomes incoherent noise rather than a recoverable scene as the number of stored patterns grows. This is the Hopfield capacity limit (catastrophic interference). Two partial fixes are: (i) *complex-valued embeddings* $\phi = e^{i\theta(\mathbf{x})} \cdot \mathrm{amp}(\mathbf{x})$, which restore a meaningful phase structure and make interference semantically interpretable; and (ii) *sparse superposition* (storing only near-orthogonal patterns), which is the CSP condition that typed basis vectors of different types do not interfere (section 2.6). Fix (ii) is a direct consequence of the CSP typed sparse structure: dimensions of different types are orthogonal by construction, so near- orthogonality of stored patterns is enforced structurally rather than as an additional regularisation. Fix (i) extends the CSP framework with a signal-theoretic technique; it is compatible with the typed sparse architecture but not derived from it.

CSP interpretation. In the CSP framework, eq. (6.13) is a global *co-occurrence metric* over the quality-dimension space: W_{ij} records how strongly dimensions i and j are jointly activated across all stored concepts. Retrieval (6.15) is the projection of a query concept onto the stored concept most similar to it in quality-dimension distance. The holographic analogy thus re-derives the core CSP operation—prototype-based nearest-neighbour retrieval—from first principles of associative memory, without requiring either a pre-specified metric or an explicit list of prototypes. The breakdown condition (inputs from unrelated semantic domains) corresponds to a CSP where the quality-dimension basis is fragmented across incompatible sub-spaces, confirming that meaningful holographic memory requires a coherent, shared quality-dimension structure.

6.24 Event-Triggered Semantic Maintenance

A CSP graph is not static. New tiddlers are inserted, concepts drift, contradictions emerge, and the digestion score (definition 2.6) of existing nodes decays as the surrounding knowledge evolves. Maintaining semantic coherence at scale requires an *event-driven scheduler* analogous to a database autovacuum or incremental compiler: background jobs triggered by graph events rather than manual curation.

Event ontology. The CSP event space partitions into five families:

Event	**Description**
`CONTENT_INSERTED`	New tiddler added to the graph
`CONTENT_MODIFIED`	Existing tiddler body or fields updated
`SIMILARITY_DETECTED`	$\delta(c_i, c_j) < \theta$ for some threshold θ
`CONTRADICTION_FOUND`	Two nodes carry incompatible Layer C assertions
`PROOF_UNDIGESTED`	SDS(c) $< \alpha$ after formalisation
`ONTOLOGY_DRIFT`	Layer A neighbourhood of c has shifted beyond tolerance
`PROOF_CLOSED`	Implication closure computation completed

Triggered maintenance jobs. Each event spawns one or more typed maintenance jobs. The mapping below lists the primary job families:

`CONTENT_INSERTED` Parse → tokenise → entity extraction → formula detection → citation linking → semantic hashing → neighbourhood search → Layer B tag assignment → SDS initialisation.

`SIMILARITY_DETECTED` Duplicate detection → abstraction candidate synthesis → merge suggestion → contradiction check → transform-reuse scan.

`CONTRADICTION_FOUND` Source attribution → Layer C consistency check → resolution proposal (retract, hedge, or fork concept) → human-review flag.

`PROOF_UNDIGESTED` Proof-to-talk transform (generate spoken narrative or slide skeleton) → pedagogical projection assignment → teach(c) update → SDS recomputation.

Periodic (nightly) Semantic vacuum: remove stale edges, recompute centrality, recompress embeddings, update Layer A neighbourhood, run implication-closure sweep (`IMPLICATION_CLOSURE`), stabilise ontology.

The scheduler can be implemented as a *Hierarchical Finite State Machine* (HFSM) over the CSP graph—the same formalism proposed in section 6.21 for expert-system processes. Each job is a typed transition with preconditions (event type and graph state) and postconditions (updated node attributes). The HFSM formulation makes job conflicts explicit (e.g., `MERGE_SUGGESTION` and `CONTRADICTION_CHECK` cannot fire simultaneously on the same node pair) and enables formal verification of scheduler liveness and safety properties.

Remark 6.2. Tao's infrastructure critique [Tao, 2026]—that faster AI tools without better knowledge infrastructure create "traffic jams" of undigested results—is precisely what this scheduler addresses. The CSP graph with its typed events and maintenance jobs is the knowledge infrastructure layer that transforms a pile of verified but opaque proofs into a navigable, digestible, reusable semantic corpus.

6.25 CSP Retrieval Algebra: Beyond Vector Similarity

Classical retrieval augmented generation (RAG) reduces every query to a vector and every document to a chunk:

$$\text{query} \;\rightarrow\; \mathbf{q} \;\rightarrow\; \operatorname{argmax}_{\mathbf{c}} \cos(\mathbf{q}, \mathbf{c}),$$

returning the nearest chunk in embedding space. This architecture conflates two distinct problems: *what to retrieve* (the semantic question) and *how to encode it* (the representation question). The result is that retrieval quality depends entirely on the quality of the embedding, the chunk boundaries are arbitrary, and the retrieved unit is a block of text rather than a typed semantic object.

The retrieval-unit mismatch. The fundamental difficulty is that the correct retrieval unit depends on the cognitive task. Answering a question about a proof step requires retrieving a *theorem* and its *dependencies*, not the surrounding paragraph. Checking a design constraint requires retrieving the relevant *constraint objects* across multiple expert domains, not the page on which the constraint was stated. The entire knowledge-retrieval industry is converging on this observation: the retrieval unit must match the cognitive task [Kolbe, 2025].

The typed sparse vector space of section 2.6 already encodes this insight: basis vectors are typed (`theorem`, `formula`, `reference`, `constraint`, `process`), and the document object ontology of section 6.20 provides the closed type system over which retrieval can be defined. What remains is to make retrieval a *typed algebraic operation* rather than a distance computation.

CSP retrieval algebra. A *CSP retrieval expression* is a term in an algebra whose primitives are typed concept filters and whose operators are semantic relations:

$$\text{retrieve}\big(\texttt{theorem},\ \text{related-to}(\texttt{formula}: X),\ \text{verified-by}(\texttt{reference}: Y),\ \text{projected-as}(\texttt{lean})\big). \tag{6.16}$$

Each term in (6.16) is a constraint on the concept graph (section 6.20): the first selects all nodes of type `theorem`; the second restricts to those connected via a `related-to` edge to the formula object X; the third further restricts to those whose citation graph includes reference Y; the fourth projects the result into the Lean type-theory representation. This is *typed compositional retrieval*: the query is a conjunction of typed graph predicates, and the result is a semantically coherent bundle of typed objects rather than a ranked list of text strings.

Projections as first-class morphisms. The operator projected-as(`lean`) in (6.16) is not a post-processing step; it is a *morphism* in the category of conceptual spaces. Each projection transforms the retrieved concept bundle into a specific representational format—Lean proof term, LaTeX equation, ScholarWiki record (section 6.26), embedding vector, natural-language paragraph—while preserving the underlying semantic relations. The CSP retrieval algebra generalises both the fibered dual (eq. (2.27)) and the multi-projection query architecture of section 6.22: the fibered dual describes which probes can be applied; the retrieval algebra describes which typed objects they retrieve; the projections describe what form the retrieved objects take.

From RAG to post-RAG architecture. The pipeline that the CSP retrieval algebra implies is:

$$\text{query} \;\to\; \text{conceptual decomposition} \;\to\; \text{typed graph traversal} \;\to\; \text{semantic bundle assembly} \;\to\; \text{projection synthesis},$$

which differs from RAG at every step. The query is decomposed into typed predicates rather than embedded as a single vector. Retrieval is graph traversal under typed constraints rather than approximate nearest-neighbour search. The result is a structured bundle of typed semantic objects rather than a bag of text chunks. And the final projection renders this bundle in whichever representational format the downstream task requires. This architecture is closer to a *semantic operating system* than a retrieval pipeline: it maintains typed concepts, provenance, constraints, and inter-concept relations as first-class runtime objects, with embeddings appearing only as one projection among many rather than as the primary storage and retrieval medium.

Transclusion and identity-preserving references. A critical failure mode of text-similarity retrieval is its treatment of *transclusion*. When Gauss's divergence theorem appears in forty papers, a vector-database system creates forty different embedding vectors for forty different text renderings. The CSP system creates *one* persistent semantic object:

$$\texttt{theorem:gauss:divergence} \;\mapsto\; \begin{cases} \pi_{\text{latex}} & \text{the \LaTeX\ source} \\ \pi_{\text{lean}} & \text{the Lean 4 proof term} \\ \pi_{\text{spoken}} & \text{the phoneme-level spoken form} \\ \pi_{\text{embed}} & \text{an embedding, as one projection among many} \end{cases}$$

Identity resolution—recognising that two surface forms refer to the same underlying object—is then a graph identity check, not a nearest-neighbour search. This matters fundamentally for mathematical and legal reasoning: a transclusion is not a quotation (same text, different origin), not a derivation (different theorem, same proof strategy), and not a paraphrase (different surface form, same meaning). Current RAG systems collapse all four relations into text similarity, destroying the provenance and identity structure that formal reasoning requires.

Transport maps as retrieval operators. The projection morphisms π_{ij} of section 2.15 are not format converters—they are *semantic transport operators* that carry conceptual structure from one type-space to another while preserving the relationships that matter for the target type. Retrieval then becomes *compositional transport*:

The spoken-form projection deserves special attention: "integral from zero to infinity" is not merely text. It preserves operator precedence, semantic grouping, and ambiguity structure as a *semantic phoneme graph*, not a character string. The projection π_{spoken} should map a formula to a graph over prosodic units, not to a plain character sequence.

$$\text{retrieve}\big(\texttt{theorem},\ \text{via} = [\pi_{\text{symbolic}}, \pi_{\text{citation}}, \pi_{\text{dependency}}],\ \text{project\_into} = \texttt{spoken\_math},\ \text{confidence} > 0.8\big), \tag{6.17}$$

where the via clause specifies which transport paths are traversed (each contributing a different distance estimate from the type-local families $\mathcal{D}_T$ of eq. (2.29)), and the project_into clause specifies the rendering morphism applied to the retrieved bundle. The confidence threshold filters the Pareto surface of multi-transport scores, retaining only concepts that are close under *all* specified transports simultaneously. This is retrieval as algebraic composition over the enriched category **CSP** (section 2.15), fundamentally unlike nearest-neighbour search in a fixed metric space.

Semantic curvature. The type-local geometry suggests a new measure of conceptual importance distinct from citation count. Define the *semantic curvature* of a concept c as:

$$\kappa(c) \;=\; \sum_{T \ni c} \sum_{\pi:\, T \to T'} \lambda_{T,T'} \cdot \left(1 - d_T(c, \partial T \mid \pi)\right), \tag{6.18}$$

where ∂T denotes the "boundary" of type T (the region of low information density), $d_T(c, \partial T \mid \pi)$ measures how far c is from the periphery under projection π, and $\lambda_{T,T'}$ are coupling weights. Informally: $\kappa(c)$ is high when c sits near the high-curvature core of *many* type-spaces simultaneously —when it is central as a theorem, central as a citation target, central as a formula dependency, and central as a diagram element. A foundational result like Nash's equilibrium theorem or Tate's thesis may have $\kappa(c) \gg 0$ despite low publication volume: it is tightly coupled to many theoretical domains across many projections. Semantic curvature is therefore a more faithful proxy for conceptual importance than bibliometric citation count, which measures social visibility rather than structural centrality in the concept graph.

CSP region algebra: projection, intersection, union. Bechberger and Kühnberger [2017] formalise three algebraic operators on CSP regions whose semantic-compiler interpretations sharpen the retrieval algebra above. The *projection operator* restricts a concept region to a subset

A of quality domains: $\pi_A(R) = \{p_A : \exists p_{\bar{A}}, (p_A, p_{\bar{A}}) \in R\}$, where p_A denotes the coordinate in the selected domains. In the retrieval expression (6.16), every typed filter is a projection: the projected-as(`lean`) clause restricts the full concept bundle to its Lean-type-theory fiber, discarding all other representational dimensions. A critical property: projection preserves star-shapedness. If R is star-shaped with prototype p^* (section 4.5), then $\pi_A(R)$ is star-shaped with prototype $\pi_A(p^*)$—semantic slicing cannot destroy prototype structure. This makes projection a safe operator for multi-stage CSP enrichment pipelines: each compilation pass of section 6.11 slices the document's semantic bundle onto a sub-domain without disturbing the prototype structure of the remaining dimensions.

The *intersection operator* fuses two concept regions under shared constraints. In the fuzzy setting, $\mu_{R_1 \cap R_2}(x) = \min(\mu_{R_1}(x), \mu_{R_2}(x))$; for star-shaped R_1 and R_2 sharing prototype p^*, their intersection is again star-shaped with respect to p^*, since the minimum of two monotone-decaying membership functions is monotone-decaying. In the CSP operator algebra of section 6.30, intersection is the reconciliation merge: two partial CSP views D_1 and D_2 of the same underlying object are fused at shared nodes via the Bayesian meet (6.23), which generalises the crisp minimum to a probabilistic meet propagating uncertainty. The *repair mechanism* of Bechberger and Kühnberger [2017]—restoring star-shapedness when intersection produces non-star- shaped fragments—corresponds to the normalisation operator (C) of section 6.30: broken chains, isolated nodes, and contradictory edges are intersection artifacts requiring local manifold smoothing to restore prototype reachability.

The *union operator* computes the smallest star-shaped superset of the union of two regions, given their shared prototype. In the retrieval algebra, union corresponds to the join $P_i \vee P_j$ of projectors (6.4): the retrieved set expands to all concepts reachable via either projector, with the closure being the minimal star-shaped region containing both projector images. α-cut reasoning then sharpens this union: the confidence filter $\mu_R(x) \geq \alpha$ (4.4) selects only concepts sufficiently central in the expanded region, suppressing spurious peripheral matches that arise from the union operation. The CSP retrieval algebra thus instantiates the Bechberger operator set as executable typed-graph operations: geometric region calculus over star-shaped manifolds becomes the algebraic semantics of the semantic compiler's retrieval and fusion passes.

6.26 *ScholarWiki: A Token-Efficient LLM Interaction Format*

The CSP retrieval algebra of section 6.25 is a *semantic* specification: it defines what typed objects are retrieved and under which graph predicates. For this algebra to be executed by a large language model—as a generator, a retriever, or a reasoning agent—the underlying semantic objects must be serialised into a concrete, machine-readable text format that is both compact enough to fit within a context window and structured enough to support typed parsing. The *ScholarWiki* format satisfies both requirements: it is a TiddlyWiki-inspired record syntax compressed to the minimum token budget compatible with full CSP semantics.

Record syntax. Each semantic unit—section, paragraph, equation, formula, citation, definition, theorem—is encoded as a single *tiddler* record consisting of a typed header line followed by an optional natural-language body:

```
--- KIND TITLE | field:value | field:value ...
body (optional)
```

where `KIND` $\in \{\textsc{sec}, \textsc{p}, \textsc{eq}, \textsc{f}, \textsc{cit}, \textsc{def}, \textsc{thm}, \ldots\}$ is the semantic type of the record (matching the typed basis of section 2.6), and `TITLE` is a hierarchical persistent identifier such as

`CSPPaper/02-Foundations/05-CategoryPerspective/para.0021`

This identifier encodes a navigational path through the document's quality domain for structure (chapter 3) and serves as the primary key for all cross-record references.

Key field vocabulary. The header fields form a typed attribute grammar over the six CSP information layers:

Field	**Meaning**	**CSP layer**
`e:`	Equation number	Structural
`lb:`	Cross-reference label	Structural
`r:`	Role (definition / lemma / theorem / approximation)	Semantic
`l4:`	Lean 4 type or proof stub	Type-theoretic (Layer C)
`f:`	Derivation sources (comma-sep. IDs)	Provenance DAG
`u:`	Downstream users (comma-sep. IDs)	Provenance DAG
`a:`	Author(s)	Bibliographic
`y:`	Year	Bibliographic
`t:`	Title	Bibliographic
`p:`	Publisher / venue	Bibliographic
`as:`	Reasoning premises	Illocutionary (Layer B)
`cl:`	Conclusion	Illocutionary (Layer B)
`imp:`	Implicit gap or epistemic hedge	Illocutionary (Layer B)

Table 6.6: ScholarWiki field vocabulary mapped to CSP information layers.

The `f:/u:` pair encodes the derivation DAG that underlies the α-cut confidence filter of eq. (4.4): the set of source IDs for a record is exactly the citation neighbourhood used by the transport retrieval expression (6.17) to propagate confidence scores backward through the dependency graph.

Transclusion syntax. Existing records are referenced via the transclusion notation

$$\texttt{[[TITLE:TYPE]]},$$

where `TYPE` is the requested projection. For example:

- `[[CSPPaper.equation.0003:EQ]]` — retrieve and display the equation object
- `[[CSPPaper.formula.0007:F]]` — retrieve the inline formula
- `[[CSPPaper.cite.gardenfors2000:CIT]]` — retrieve the citation record

This is the concrete realisation of the projection morphisms introduced in section 6.25: `[[X:T]]` is syntactic sugar for the typed retrieval expression retrieve(T, id $= X$). Because each object has a persistent identifier decoupled from its surface rendering, the same theorem can be transcluded as a LaTeX equation, a Lean proof term, a spoken-math phoneme graph, or an embedding vector—the four projection targets of section 6.25— without creating redundant copies. Identity resolution is a string comparison on `TITLE`, not nearest-neighbour search.

This eliminates the transclusion failure mode described in section 6.25: Gauss's divergence theorem appearing in forty papers maps to one record `theorem.gauss.divergence` with forty transclusion references, not forty independent embedding vectors.

Advantages for LLM interaction. Four properties make ScholarWiki preferable to both raw LaTeX and chunk-based RAG for LLM-based processing:

Token efficiency. The header line is maximally compact: a complete theorem record with provenance, Lean stub, and three field annotations fits in ≈40–80 tokens. Repeated full path strings are replaced by short IDs in the `f:/u:` fields.

Structured generation. An LLM generating new tiddlers fills a fixed syntactic template rather than producing free-form prose. This constrains output to the typed CSP algebra and enables deterministic parsing without heuristic post-processing.

Verifiable provenance. The `f:/u:` DAG is machine-checkable: every claim can be traced to its source records. This is the document-level analogue of the Lean proof-term chain (section 6.11).

Multi-projection semantics. A single record (e.g., the Gaussian splatting equation (6.21)) can be retrieved as EQ (LaTeX source), F (inline fragment), or via `l4:` as a Lean statement, with each projection computed on demand rather than stored separately.

Connection to the retrieval algebra. Every tiddler header is a *pre-parsed retrieval expression*: the `KIND` field corresponds to the type filter in (6.16); the `lb:` field provides the cross-reference hook used by typed graph predicates; `as:/cl:/imp:` encode the Layer B illocutionary annotation of section 2.7, enabling the CSP-to-CSP operator pipeline of section 6.30 to operate directly on ScholarWiki records without an intermediate parsing pass. The full manuscript encoded in ScholarWiki format is thus itself a *CSP semantic graph*: vertices are tiddler records, edges are transclusion links, and the retrieval algebra operates over this graph natively.

Remark 6.3 (Semantic copy-paste via drag-and-drop). A TiddlyWiki-based ScholarWiki implementation supports *drag-and-drop transfer* of tiddlers between wiki instances. Unlike HTML copy-paste, which copies surface text and breaks all typed relations, TiddlyWiki drag-and-drop preserves the full tiddler structure: `KIND`, `TITLE`, all field-value pairs, and transclusion links. A tiddler dragged from one project wiki to another arrives as a typed semantic object with its provenance (`f:/u:` DAG), Lean stub (`l4:`), and citation links intact. Missing dependencies appear as broken transclusion links, which fire `CONTENT_MODIFIED` events in the maintenance scheduler (section 6.24), triggering namespace-remapping and backlink-repair jobs automatically. This makes distributed collaborative curation—Tao's Polymath scenario—low-friction in a way unavailable to standard blogs, document stores, or vector databases [Tao, 2026].

6.27 Commutative Diagrams as CSP Objects

Among all mathematical objects that appear in scientific documents, the *commutative diagram* occupies a unique position: it is simultaneously a geometric object, a logical statement, a computational specification, and a visual representation. This makes it a near-perfect CSP object—an entity that inhabits multiple quality-dimension spaces simultaneously and admits projections to each.

The five-fold semantic structure. A commutative diagram $D = (V, E, \lambda, \sim)$ encodes:

1. *Entities*: the nodes V (objects in a category);
2. *Morphisms*: the labelled arrows E with $\lambda\colon E \to \mathsf{Hom}$ (functions or transformations between objects);
3. *Constraints*: the commutativity conditions $\sim$ asserting that any two directed paths between the same endpoints compose to the same morphism;
4. *Equivalences*: the isomorphisms among objects (arrows with two-sided inverses), which induce HoTT-style identifications (section 2.12); and
5. *Compositionality*: the composition law $f \circ g$ as the fundamental operation, making every diagram a fragment of a larger algebraic structure.

All five components are directly expressible as typed entries in the sparse vector representation of section 2.6: nodes as `entity`-typed vectors, arrows as `pointer`-typed vectors, commutativity conditions as `theorem`-typed constraint vectors, and isomorphisms as nilpotent pairs $e_i \wedge e_j = 0$ (eq. (2.8)).

Multi-projection architecture for diagrams. Because a commutative diagram already encodes its semantic content structurally, it admits natural projections to multiple representational formats:

- π_{tikz}: the `tikz-cd` source code rendering the diagram as a LaTeX figure;

- π_{graph}: a directed graph (JSON or DOT) for visualisation via Graphviz, Mermaid, or Cytoscape;

- π_{lean}: a Lean 4 theorem asserting the commutativity condition, checkable by the Mathlib type-checker [The mathlib Community, 2020];

- π_{embed}: a vector embedding of the diagram's semantic content for similarity search; and

- π_{csp}: the CSP sub-graph of entities and constraint edges, pluggable directly into the retrieval algebra of section 6.25.

The commutative diagram is thus the archetypal multi-projection concept: a single semantic object with five natural projections, each useful for a different downstream task. This makes it the natural *test object* for any CSP pipeline—a system capable of handling diagrams correctly handles the full complexity of typed, multi-representational, constraint-bearing semantic objects.

6.28 Scientific Semantic Reconstruction

A STRUCTURAL PARALLEL EXISTS between internet-scale 3D scene reconstruction and CSP semantic manifold reconstruction that illuminates both fields. The computer-vision community spent fifteen years solving the *Structure from Motion* (SfM) problem: given millions of heterogeneous, partially overlapping images of the world, infer the latent 3D geometry that explains all observations consistently. The resulting architecture—keypoint matching, pose-graph construction, and global bundle adjustment—is not specific to vision; it is a general solution to the problem of *latent object reconstruction from partial, noisy, multi-view observations.*

Scientific knowledge has the same structure. Papers are partial, noisy observations of an underlying latent conceptual manifold. The geometry of that manifold is not observed directly; it must be inferred from consistency constraints across overlapping observations.

The SfM–CSP correspondence. The mapping between vision and knowledge is structurally precise:

3D Vision (SfM)	CSP Semantic Reconstruction
Image (partial view of scene)	Document (partial view of concept manifold)
SIFT keypoint	Formula AST node / theorem anchor
Camera pose	Source viewpoint: bias, era, notation style
Scene point (3D coordinate)	Abstract concept (CSP coordinate)
Epipolar constraint	Citation / transclusion consistency
Bundle adjustment	Global semantic consistency optimization
Long-tail reconstruction	Rare concept recovery (low-frequency symbols)
Multi-view stereo	Multi-source semantic triangulation
Depth prior	Formal hierarchy prior (type system)
Photometric consistency	Symbolic equivalence under projection
Scene reconstruction	Conceptual-space manifold inference

Multi-view latent objects. A single semantic object—a theorem, a formula, a concept—appears in many documents as many surface forms: LaTeX source, OCR text, spoken utterance, handwritten derivation, citation key, textbook paraphrase, Lean proof term, computer-algebra AST. These are not different objects; they are *different camera views* of the same hidden semantic object. The analogy is exact: just as two photographs of the same building from different angles encode the same 3D structure through different 2D projections, two renderings of the same theorem encode the same conceptual object through different notation projections π_α (section 2.10). This motivates a *multi-view reconstruction* framing: given k projections $\pi_\alpha(F)$ of an unknown concept F, recover F itself.

$$\hat{F} = \arg\min_{F \in \mathcal{M}} \sum_{\alpha=1}^{k} \left\| \pi_\alpha(F) - x_\alpha \right\|_{S_\alpha}^2, \qquad (6.19)$$

where x_α is the observed projection, $\|\cdot\|_{S_\alpha}$ is the metric of subspace S_α, and minimisation is over the formula manifold $\mathcal{M}_{\text{formula}}$ of section 2.10. Each observation x_α is a noisy, potentially partial rendering; recovery requires combining multiple views.

Long-tail collapse and popular-concept bias. The SfM community discovered that naïve feature matching works well for famous landmarks (the Eiffel Tower, the Coliseum) because thousands of overlapping images provide redundant constraints, but fails for the "long tail" of ordinary locations photographed by only one or two people. The analogous failure mode in CSP is *popular-concept collapse*: dense embedding models fine-tuned on internet-frequency corpora overfit to dominant symbolic vocabularies and lose resolving power for rare concepts—obscure notation, niche terminology, handwritten derivations, low-citation bridge theorems. Standard cosine similarity cannot distinguish a highly-cited well-known theorem from a subtle but structurally identical variant that appears in only two papers.

The SfM solution used geometric priors, cross-scene transfer, and synthetic augmentation. The CSP equivalents are:

- *Formal priors*: type-system constraints (section 2.15) that regularise rare-concept positions via their type neighborhood.
- *Cross-domain transfer*: projection morphisms that carry geometric structure from rich type-spaces (e.g. theorem-dense domains) to sparse ones.
- *Synthetic symbolic augmentation*: equivalent reformulations of rare formulas generated by computer-algebra systems to increase observational coverage.

Semantic bundle adjustment. Bundle adjustment is the keystone of global SfM: rather than solving camera poses and scene point positions sequentially, it optimises all variables jointly subject to all reprojection constraints simultaneously. The result is orders-of-magnitude more accurate than sequential estimation because errors cannot propagate unchecked along a fixed reconstruction order.

The CSP equivalent is *semantic bundle adjustment*: jointly optimise all concept coordinates $\{c_k \in \mathcal{M}\}$ subject to all available consistency constraints:

$$\min_{\{c_k\}} \sum_{\text{citations}} w_{\text{cit}}\, \delta(c_i, c_j)^2 + \sum_{\text{transclusions}} w_{\text{tc}}\, \|\pi(c_i) - c_i'\|^2 + \sum_{\text{Lean eqs.}} w_L\, \|c_i - c_j\|^2 + \sum_{\text{contradictions}} w_\perp\, \max(0,\, m - \delta(c_i, c_j))^2, \tag{6.20}$$

where the first three terms are attractive (citation dependency, transclusion overlap, Lean-verified equivalence encourage proximity) and the fourth term is repulsive (contradictions enforce a minimum separation m in semantic space). Implementation candidates include factor graphs, belief propagation, graph-neural energy minimisation, or sparse Levenberg–Marquardt (the method used in SfM) applied to the concept graph.

The repulsive contradiction term makes CSP geometry distinct from all standard embedding losses: it encodes an epistemic force field. A concept contradicted by a well-established theorem cannot sit near it in the manifold.

Contradiction geometry. Equation (6.20) introduces a dichotomy of semantic forces that standard similarity retrieval ignores:

Relation	**CSP Force**
`supports`	Attractive — pull concepts together
`contradicts`	Repulsive — enforce minimum separation
`generalises`	Containment — one convex region contains another
`equivalent`	Identity collapse — manifold merge
`approximates`	Soft overlap — fuzzy containment

Encoding contradictions as geometric repulsion creates *semantic force fields* inside the manifold. A coherent research front forms a cluster of nearby, mutually supportive concepts. A paradigm conflict appears as two clusters with a repulsive boundary between them. Detecting an emerging consensus corresponds to watching two previously separated clusters converge as new bridge theorems reduce the contradiction tension.

Semantic SLAM. Simultaneous Localisation and Mapping (SLAM) is the robotics discipline of building a map of an unknown environment while simultaneously tracking position within it. The CSP analogue is *semantic SLAM* for scientific knowledge:

new document $\rightarrow$ local semantic map $\rightarrow$ align with global CSP $\rightarrow$ update global manifold $\rightarrow$ propagate revised coor

Each newly ingested paper is first placed locally (which existing concepts does it reference? which new concepts does it introduce?), then its novel concepts are globally aligned with the existing manifold via (6.20), and finally the manifold is updated by propagating the new geometric constraints. This incremental architecture avoids full re-optimisation on each ingestion: local loops converge fast; global bundle adjustment runs periodically in a background pass.

Gaussian semantic splatting. Three-dimensional Gaussian Splatting represents scenes as sets of anisotropic Gaussian primitives, each with a position, covariance, opacity, and appearance. The rendering equation accumulates contributions from nearby Gaussians along each query ray. The CSP analogue represents each semantic fragment as a *Gaussian semantic splat*:

$$G_k = \big(\mu_k,\, \Sigma_k,\, \alpha_k,\, \tau_k,\, \mathbf{v}_k\big), \tag{6.21}$$

where $\mu_k \in \mathcal{M}$ is the latent semantic centre, Σ_k is an uncertainty tensor over quality dimensions, $\alpha_k \in [0,1]$ is the confidence (opacity analogue), τ_k is the temporal coordinate (era of origin), and $\mathbf{v}_k$ is the typed sparse vector of section 2.6. A semantic query accumulates splat contributions weighted by proximity and confidence, analogously to alpha-compositing in Gaussian splatting. This formulation naturally handles uncertainty, temporal drift, and sparse observations: concepts from one paper are represented as wide Gaussians that narrow as more observations arrive, exactly as photometric constraints sharpen 3DGS reconstructions.

Bibliography as multimodal sensor fusion. Every scientific bibliography aggregator provides different signal about the same underlying semantic objects. Their combination is a *multimodal sensor fusion* problem, structurally identical to fusing RGB cameras, LiDAR, and GPS in autonomous driving:

Bibliography source	**Primary signal**
Crossref	DOI identity and metadata authority
arXiv	Fulltext and open-access mathematical content
Semantic Scholar	Citation-graph enrichment and influence scores
OpenAlex	Graph-topology statistics and institution provenance
DBLP	Computer-science normalisation and authorship
Lean / Mathlib	Formal proof structure and theorem dependency

Each source is a noisy sensor with known biases. Crossref is authoritative on identity but has no semantic content. arXiv has richest mathematical content but limited citation data. Semantic Scholar has citations but no fulltext. Combining these via (6.20)—letting each source contribute its constraints—produces a semantic manifold estimate more accurate than any single source alone, just as sensor fusion outperforms any individual modality.

Citation saturation and scale-free structure. Empirically, most scientific domains saturate semantically after a few thousand important papers: the citation graph has a power-law degree distribution, most semantic mass concentrates in a core of $O(10^3)$ papers, and additional ingestion beyond this core adds marginal new constraints. This mirrors the scale-free structure of 3D internet photo collections, where most semantic mass is concentrated in famous landmarks and the long tail adds low-information redundancy.

The implication is that retrieval quality dominates corpus size past a saturation threshold. A corpus of 3 000 carefully curated, richly annotated papers with full semantic decomposition (theorems, proofs, equations, citations, formal equivalences) will outperform a corpus of 300 000 papers processed as plain-text embeddings. The CSP architecture is designed precisely for this regime: small enough for full global optimization, rich enough to benefit from it.

Saturation also defines a practical research programme: identify the core of $O(10^3)$ foundational papers in each domain, perform full semantic decomposition on this core, and use the resulting high-confidence manifold as the anchor for sparse extension to the long tail.

6.29 Large-Scale Implication Closure: A CSP Case Study

TAO'S EQUATION THEORIES PROJECT [Tao, 2026] provides a concrete large-scale validation target for the CSP implication-graph architecture. The project resolved 22 million equational implications among single-operation algebraic laws using a mix of automated provers and targeted human input: approximately 99% of cases were closed by automation, with the remaining 1% requiring mathematical insight that directed the automated search.

CSP interpretation. The project exhibits precisely the features that the CSP framework is designed to handle:

- *Modularity*: each law is an atomic statement with a typed identifier and a dependency set—a `theorem`-typed CSP node with explicit `f:` edges.
- *Clear metric*: the implication relation ("law A implies law B") is a typed directed edge with a binary truth value, making the closure computation a typed graph reachability query.
- *Mixed automation*: the 99% automated fraction corresponds to CSP operators (A)–(E) of section 6.30; the 1% human fraction corresponds to operator (F) (temporal/human-in-the-loop enrichment).

- *Implication closure*: computing the transitive closure of the implication graph is an `IMPLICATION_CLOSURE` maintenance job in the event scheduler of section 6.24, executable as a periodic background task over the CSP graph.

The generalisation from binary implication to the full typed CSP relation algebra (defines, uses, derives, contrasts, recontextualises) makes the CSP implication graph a strict extension: the Equation Theories closure is the degenerate case where all typed edges reduce to a single binary relation.

Digestion bottleneck. After closure, the 22 million results were correct but largely opaque—precisely Tao's digestion problem. The `PROOF_UNDIGESTED` event of section 6.24 would fire for every newly closed implication whose SDS (eq. (2.10)) falls below threshold, spawning proof-to-talk transforms and pedagogical projection jobs. The Equation Theories corpus thus becomes a concrete benchmark for the Digestion Layer D: measure SDS across the corpus before and after applying the maintenance scheduler.

Polymath and collaborative modular mathematics. Tao's Polymath projects succeeded when problems were decomposable into modular sub-problems with clear metrics and low coordination cost; they faltered when sub-results needed complex manual integration [Tao, 2026]. The CSP backend addresses the coordination bottleneck directly:

- *Semantic persistence*: partial results from multiple contributors are stored as typed tiddlers with provenance (`f:/u:` fields), preventing the "everyone restates everything" pathology of long blog threads.
- *Automatic abstraction extraction*: `SIMILARITY_DETECTED` events trigger abstraction candidate synthesis, surfacing patterns across contributions without manual curation.
- *Transclusion*: lemmas proved in one sub-project are transcluded into dependent sub-projects via `[[ID:THM]]` references, maintaining identity-preserving links rather than copy-pasted text.
- *Event-based task assignment*: a `CONTRADICTION_FOUND` event automatically flags a new sub-problem and notifies relevant contributors.

A CSP-backed Polymath instance would close the gap between the computational power of automated provers and the organisational intelligence of a distributed human research community—making Tao's vision of AI as a collaborative mathematical infrastructure concrete and operational.

6.30 CSP-to-CSP Operators: An Algebra of Graph Enrichment

THE SEMANTIC COMPILER PIPELINE described in section 6.11 treats documents as *inputs* to a CSP construction process. But once a CSP graph exists—however partially—the more powerful operation is *CSP-to-CSP transformation*: iteratively enriching, reconciling, and normalising the graph without returning to raw text.

Every format adapter (LaTeX parser, TiddlyWiki extractor, BibTeX expander, LLM annotation call) produces a *partial CSP view*—a noisy, incomplete observation of the same underlying semantic object. The correct framing is therefore not a pipeline but a *posterior inference problem*:

$$\text{CSP} = \arg\max_G P(G \mid \text{LaTeX AST, TW JSON, BibTeX, LLM annotations}, \ldots), \tag{6.22}$$

where G ranges over typed semantic graphs and each format contributes its own projection of the same document. Two partial CSP views D_1, D_2 are merged not by concatenation but by Bayesian fusion:

$$P(H \mid D_1, D_2) \;\propto\; P(D_1 \mid H)\, P(D_2 \mid H)\, P(H), \tag{6.23}$$

where H is the latent graph hypothesis. Each format channel is a noisy annotator; the fused CSP carries explicit uncertainty over every node and edge, which is essential for multi-LLM ensemble fusion and for multi-format alignment.

The Bayesian framing of eqs. (6.22) and (6.23) makes the CSP immune to the silo pathology of classical MBSE: each new format is a new sensor adding evidence, not a competing representation requiring manual reconciliation.

Operator taxonomy. Six families of CSP-to-CSP operators cover the full enrichment lifecycle:

(A) Enrichment operators Add structure inferred from auxiliary signals. *Structural completion* infers missing clause nodes, dependency edges, and reference targets. *Type refinement* promotes coarse tags (NOUN → ENTITY/PROCESS/MEASURE/SYMBOL). *Attribute augmentation* attaches confidence, provenance, span links, embeddings, and projection scores.

(B) Reconciliation operators *Node alignment* merges duplicate representations of the same entity across sources (e.g. the same reference appearing in raw text, in a BibTeX record, and in an LLM expansion) into one canonical node with alias sets and a confidence distribution. *Edge conflict resolution* merges or hierarchises contradictory relation labels. *Cross-view fusion* aligns the topologies of distinct format projections into the shared CSP graph.

(C) Normalisation operators *Canonicalisation* maps synonyms, formatting variants, and identifier styles to a unified identity space. *Granularity harmonisation* aligns nodes at different levels (document, section, paragraph, sentence, clause, token/entity) into a consistent hierarchy with permitted cross-level edges.

(D) Probabilistic operators Every node carries $P(\text{node exists})$, $P(\text{type} = X)$, and $P(\text{edge exists})$; these are propagated and never collapsed prematurely. Ensemble voting and semantic consensus emerge from accumulated probability mass across multiple projections.

(E) Embedding operators After structural enrichment, recompute node vectors and manifold coordinates: *graph re-embedding* updates each concept's latent position given its new neighbourhood; *local manifold smoothing* repairs broken chains, isolated nodes, and contradictory edges.

(F) Temporal operators Track the evolution $\text{CSP}_t \to \text{CSP}_{t+1}$ by computing the *CSP diff*: added/removed nodes, changed edges, shifted confidence values. This is essential for iterative refinement loops and versioned TW2TW-style knowledge-graph evolution.

Semantic reconstruction as a quality metric. The operator algebra enables a projection quality check. After applying an operator sequence, regenerate a paragraph from the CSP and compare it against the source:

$$L_{\text{semantic}} = d(D, \hat{D}), \tag{6.24}$$

where D is the source document and $\hat{D}$ the regenerated version. Low L_{semantic} indicates that the CSP captures the document faithfully; high values flag missing nodes or broken edges. The architecture is no longer a parser: it is a *semantic compiler with iterative belief refinement over a typed graph substrate*, with the CSP serving as the intermediate representation (IR) throughout.

D_{CSP}*: a multi-type evaluation distance.* The enriched category **CSP** naturally yields one distance per type family $\mathcal{D}_T$ (eq. (2.29)). Aggregating these into a single reconstruction quality score gives the *CSP evaluation distance*: let G_{ref} be a reference CSP graph (e.g. a human-annotated gold standard or a Lean-verified core). For a generated graph G, define

$$D_{\text{CSP}}(G_{\text{ref}}, G) \;=\; \sum_i \alpha_i \, \Delta_{T_i} \;+\; \beta \, \Delta_{\text{glue}}, \tag{6.25}$$

where Δ_{T_i} is the Hausdorff distance between the typed concept regions of type T_i in G_{ref} and G (measuring per-type reconstruction fidelity), and Δ_{glue} penalises violations of the sheaf gluing condition— cases where the projections of the same concept onto different type-spaces are mutually inconsistent. The weights α_i and β are learnable and can be calibrated against the formal component of each type: α_{theorem} can be set by Lean compiler success, α_{equation} by CAS verification. Equation (6.25) subsumes L_{semantic} as its text-domain term and extends it with per-type fidelity checks and cross-type consistency—the CSP analogue of the multi-view D_{total} metric proposed for multi-layer verification pipelines.

6.31 Engineering Systems as Latent Semantic Objects

THE MOST DEMANDING TEST of the CSP framework is whether it scales to real *engineering systems*—physical products with PCB assemblies, firmware revisions, assembly drawings, bills of materials, FMEA tables, simulation models, burn-in logs, and field-failure reports. The central claim is that all of these artefacts are not primary objects but *projections* of one evolving latent engineering object-space: applying the CSP decomposition of section 6.30 to the engineering domain.

Engineering artefacts as coordinate projections. Every engineering document corresponds to a specific projection operator applied to the latent product graph:

Artefact	**CSP Projection**
FMEA table	Failure-risk projection
Bill of materials	Procurement / materialisation projection
STEP / CAD model	Geometric projection
PCB netlist	Electrical-topology projection
SPICE model	Analog-behavioural projection
Modelica model	Multi-domain dynamical projection
Assembly drawing	Manufacturing-sequence projection
Firmware repository	Executable-behaviour projection
Burn-in report	Reliability-trajectory projection
Requirements document	Constraint projection
Service manual	Maintenance-action projection

This reframes the classical model-based systems engineering (MBSE) synchronisation problem. The failure mode of traditional MBSE is that CAD $\neq$ FMEA $\neq$ requirements $\neq$ firmware $\neq$ BOM $\neq$ simulation $\neq$ field data, and humans manually maintain the mappings. The CSP reinterpretation resolves this architecturally: *all artefacts are lossy views of one latent engineering state.* Synchronisation becomes a projection-consistency problem governed by eq. (6.20), not a manual data-entry task. The emerging Semantic Digital Thread literature moves in this direction, replacing tool-centric MBSE with semantic-data-centric engineering via knowledge graphs and federated engineering data; the CSP view goes further by replacing discrete ontological schemas with a continuous latent manifold whose coordinates encode physical mechanisms, load paths, and causal chains.

Hidden requirements as negative space. A mechanical drawing is not geometry. It is *compressed engineering intent*: an M5 chromium-coated screw drawn at a particular mounting point may implicitly encode preload force, vibration resistance, fatigue life, galvanic compatibility, assembly torque, thermal-cycling range, salt-fog resistance, EMC grounding continuity, fire certification, and a specific locking method—almost none of which is written explicitly. When a company-wide materials substitution (e.g. replacing chromium-coated screws with stainless steel to meet an environmental directive) is applied without querying this latent constraint space, the result can be catastrophic: stainless steel cannot deliver the required preload under dynamic vibration and thermal cycling, yet the requirement was never written. It existed only in old engineers' notes, obsolete test reports, certification discussions, and tribal memory.

The CSP insight is that *requirements are often negative space*—not "use chromium coating" but "maintain minimum preload under thermal cycling and vibration", with the original material as a historical witness. A CSP that tracks component ↔ physical mechanism ↔ load path ↔ environment ↔ certification rationale ↔ historical failure ↔ maintenance behaviour as a multi-layer causal graph makes this dependency visible and survivable across design iterations.

LLM as latent requirement excavation. The most valuable near-term role for LLMs in engineering is not design automation but *latent requirement excavation*: given a drawing, a BOM, a material database, a historical FMEA, and field-failure logs, generate the missing engineering questions that the artefacts do not state. For the screw substitution case, a constrained projector (section 2.8) would generate: Is the preload force safety-critical under vibration? Does the torque specification assume a specific coating friction coefficient? Was fatigue testing performed with the original screw type? Is grounding continuity required through the fastener? The system generates *questions*, not answers—a safer and more realistic role that directly prevents catastrophic implicit-requirement loss.

Temporal engineering semantics. Engineering truth is time-dependent. The same component may carry different dominant semantic coordinates across decades: a screw that is first identified as a structural member, later as a corrosion-critical interface, then as an RoHS compliance problem, then as a single-supplier risk, and finally as a cost-reduction target. Each era imposes a different projection emphasis on the same physical object. Static ontologies cannot represent this; the temporal trajectory $c(t) \in \mathcal{M}(t)$ of chapter 7 does. The *Semantic Digital Thread*—provenance of every design decision across the product lifecycle—is the engineering analogue of the citation graph in scientific CSP, and the temporal operator (F) of section 6.30 provides its formal infrastructure.

FMEA as a CSP subspace. An FMEA row already implicitly constructs a local semantic subgraph: component → failure mode → mechanism → system effect → detection method → mitigation → manufacturing process → maintenance behaviour → safety severity. Traditional FMEA is painful because engineers maintain consistency manually across all columns. In CSP terms, the FMEA is a *failure-risk projection* of the product graph. Nearest-neighbour retrieval in the latent product space—over geometry similarity, thermal profile, assembly-process history, and supplier identity—could cluster failure modes geometrically and predict latent failure risk for new designs by analogy with historical field failures. This is a *failure-space retrieval algebra* directly matching the retrieval algebra of section 6.25, and constitutes a concrete evaluation target for the engineering CSP hypothesis.

6.32 Energy-Based Models and the CSP Energy Landscape

The connection between conceptual spaces and *energy-based models* (EBMs) is deeper than analogy. Both frameworks evaluate *global compatibility* of a candidate state; both prefer states that simultaneously satisfy many coupled constraints; and both frame reasoning as optimisation over a structured latent space rather than sequential token generation.

EBMs as global constraint satisfaction. An energy-based model assigns a scalar score to a complete candidate state:

$$E(x) \in \mathbb{R}, \tag{6.26}$$

where lower energy indicates higher compatibility with the problem's constraints. Inference is optimisation: $x^* = \arg\min_x E(x)$. The contrast with autoregressive generation is exact. A language model predicts tokens $t_1, t_2, \ldots, t_n$ sequentially; an EBM evaluates whether a complete configuration x satisfies all constraints simultaneously. Hard reasoning problems—constraint satisfaction, formal proof search, engineering design verification—are fundamentally optimisation problems, not generation problems, and are therefore better served by EBMs than by next-token prediction alone [LeCun, 2022].

CSP as a multi-scale constrained energy landscape. The CSP is not a vector database. It is a *multi-scale constrained energy landscape* whose low-energy configurations are globally consistent semantic states. The compiler pipeline of section 6.11 is not extraction but iterative energy minimisation: each operator pass of section 6.30 reduces the total inconsistency energy of the graph. Four levels structure the landscape:

Level 0 — Raw observations. PDF glyphs, layout fragments, OCR output, diagrams, CAD metadata, simulation traces. Energy measures observation likelihood: how well does the raw signal match known symbolic patterns?

Level 1 — Symbolic entities. Formulae, materials, dimensions, theorems, procedures, constraints, standards, references. Energy measures internal consistency of each extracted entity.

Level 2 — Local energy functions. Each concept defines local compatibility functions $E_i(x)$: unit consistency, dimensional correctness, material compatibility, theorem applicability, parser confidence, citation reliability, topology validity.

Level 3 — Global consistency. The total energy over the entire CSP graph:

$$E_{\text{total}}(x) = \sum_i \lambda_i \, E_i(x) + \sum_{i<j} E_{ij}(x_i, x_j), \tag{6.27}$$

where λ_i weights each local energy and E_{ij} penalises inconsistency between adjacent concept pairs. Semantic bundle adjustment (eq. (6.20)) is an instance of this global minimisation.

VQ-VAE constraint codebooks. Vector-quantised variational autoencoders (VQ-VAEs, chapter 3) provide a natural interface to EBMs: the VQ-VAE compresses continuous observations to a discrete codebook $\mathcal{C} = \{e_k\}_{k=1}^{K}$, and the EBM operates over the quantised code $z_q \in \mathcal{C}$ rather than the full continuous latent space. Each codebook entry becomes a *constraint codebook*: a cluster of semantically compatible states the EBM evaluates efficiently. This reduces the search space from a continuous manifold to a finite codebook, making global minimisation of E_{total} tractable on commodity hardware for corpora of the $O(10^3)$-paper scale identified in section 6.28.

JEPA and latent predictive structure. LeCun's Joint Embedding Predictive Architecture (JEPA) provides a related framing: rather than reconstructing raw observations, the system predicts abstract latent representations of missing content [LeCun, 2022]. Applied to CSP, JEPA suggests that the compiler should predict *missing concept positions* in the manifold—not reconstruct missing text—given the observed neighbourhood. This aligns precisely with the multi-view reconstruction objective of eq. (6.19): both recover a hidden latent object F from partial projections. The EBM evaluates whether the predicted position is globally consistent; JEPA provides the prediction candidate.

Hierarchical JEPA (H-JEPA) introduces a multi-scale prediction hierarchy that maps onto the four energy levels above: each JEPA layer predicts the abstract representation that the next higher level will evaluate for consistency.

Layered reasoning stack. The practical consequence is a three-tier architecture for knowledge-intensive scientific and engineering tasks:

1. **LLMs** for candidate generation and natural-language communication: propose concept positions, generate hypotheses, expand citations into BibTeX, translate drawing annotations into structured queries.

2. **EBMs** for constraint-heavy reasoning: evaluate global compatibility of candidate CSP states, detect constraint violations, guide search toward low-energy configurations—precisely the role played by the operator families of section 6.30.

3. **Formal verifiers** (Lean / Mathlib for mathematics; standards checkers for engineering) for correctness certification: a Lean proof that compiles is correct by construction; an EBM configuration satisfying all engineering constraints is nominally design-valid.

Recent formal reasoning systems demonstrate that EBM-style constraint evaluation combined with Lean verification can achieve near-complete coverage on benchmark theorem corpora where autoregressive LLMs plateau well below 50%. The CSP framework is an unusually suitable substrate for this stack: it is a structured state space with coupled constraints, multiple abstraction levels, and nontrivial latent variables—exactly the regime where EBMs complement rather than compete with LLMs.

Such results are reported on company benchmarks and should be read as directional evidence rather than independently verified conclusions; formal correctness guarantees hold relative to the specification, not to real-world intent.

6.32.1 Selective Semantic Crystallization: Lessons from RecMem

RecMem [Liang et al., 2024] is a recurrence-based conversational memory architecture that consolidates chat history via similarity and frequency thresholds: observations accumulate in a latent buffer and are promoted to semantic memory only when they recur above a configurable count. The system is operationally clever— economical, asynchronous, and effective for chat-agent contexts—but rests on a fundamentally embedding-centric, text-oriented model of memory.

The CSP architecture has a different target: persistent scientific knowledge with typed relational structure, formal morphisms, and algebraic reconstruction. The mismatch is worth stating precisely: *RecMem optimises conversational memory via recurrence thresholds; CSP targets durable scientific semantics via structured relational topology (symbolic + geometric + graph).* Nevertheless, RecMem offers genuine heuristics for *when* and *how* to delay semantic stabilisation, and its maintenance scheduling validates intuitions already built into the CSP event-trigger framework (section 6.24).

Concept mapping. Section 6.32.1 reinterprets RecMem's components within the CSP compiler.

RecMem concept	**CSP reinterpretation**
Subconscious buffer	Unstable semantic fragments (pre-crystallization objects)
Episodic memory	Stabilised process narratives (Layer B typed events)
Semantic memory	Canonical concept nodes (typed sparse basis elements)
Recurrence threshold	Graph/topology reinforcement count
Similarity threshold	Morphism compatibility distance δ_T
Consolidation	Semantic crystallisation (canonical-object promotion)
Vector clustering	Typed graph emergence from fragment coalescence

Table 6.7: RecMem concepts reinterpreted as CSP compiler constructs.

Delayed consolidation and selective crystallisation. RecMem's key insight is that many observations are noise until recurrence proves stability. Translated into the CSP setting, this becomes the principle of *selective semantic crystallisation*: the compiler must avoid premature canonicalisation, ontology freezing, or irreversible commitment to a concept node. A newly ingested theorem sketch is not immediately promoted to a canonical `theorem`-typed CSP object; instead, it occupies the *latent semantic buffer*—a weakly typed, tentatively linked sub-graph—until topological and structural reinforcement justify promotion. This prevents notation-variant duplicates from spawning independent canonical nodes and avoids ontology inflation in corpora with overlapping derivations.

We adopt RecMem's staged stabilisation but replace pure recurrence with structural, symbolic, and topological reinforcement:

Multidimensional importance: beyond recurrence. RecMem uses frequency as the primary importance signal, which fails for scientific cor-

Pipeline stage	CSP meaning
Raw ingest	OCR / MathPix / LaTeX-to-AST fragments
Latent semantic buffer	Unresolved semantic objects; no canonical IDs assigned
Weakly linked graph	Tentative typed edges; low confidence Layer B tags
Recurrence reinforcement	Repeated structural motifs cross-document
Consolidation trigger	Multidimensional promotion candidate (see eq. (6.28))
Stabilised concept node	Canonical semantic object; persistent ScholarWiki tiddler
Algebraic integration	CSP manifold embedding; SDS (eq. (2.10)) computed

Table 6.8: Multi-stage semantic stabilisation pipeline, adapted from RecMem's consolidation model and extended with CSP-specific structure.

pora: a theorem may appear exactly once yet dominate the dependency structure of an entire field. A foundational lemma cited by a hundred downstream proofs is critical; a frequently repeated boilerplate acknowledgement is not.

We therefore replace recurrence-only importance with a composite metric. For each concept vertex v in the CSP graph, define

$$I(v) \;=\; \alpha\, R(v) + \beta\, C(v) + \gamma\, D(v) + \delta\, T(v) + \epsilon\, G(v), \qquad (6.28)$$

where the five components are:

$R(v)$ **Recurrence.** Cross-document co-occurrence count (RecMem's original signal); relevant but not sufficient.

$C(v)$ **Graph centrality.** Betweenness or eigenvector centrality in the typed citation/dependency graph; identifies structural hubs.

$D(v)$ **Dependency depth.** Number of canonical concepts that depend on v; deep-dependency nodes are irreplaceable.

$T(v)$ **Transformation reuse.** Number of distinct operator passes (the six families of section 6.30) that successfully apply to or through v; measures semantic generativity.

$G(v)$ **Semantic generativity.** Number of distinct new concepts whose derivation graph passes through v; analogous to the out-degree of a mathematical axiom.

The weights $(\alpha, \beta, \gamma, \delta, \epsilon)$ are learnable and can be calibrated against ground-truth importance ratings or proxy signals such as Lean verification success and SDS (eq. (2.10)).

RecMem-inspired maintenance jobs. Section 6.32.1 extends the event-trigger job taxonomy of section 6.24 with RecMem-inspired triggers specific to scientific corpora—richer than RecMem's single similarity/count gate:

Beyond RecMem: adaptive semantic annealing. RecMem uses fixed similarity and count thresholds throughout a session. A more powerful approach for evolving scientific corpora is *semantic annealing*: adaptive thresholds that begin permissive (low barriers to promotion, conservative ontology) and harden over time as the corpus matures (stricter morphism compatibility, irreversible canonical assignments). Concretely:

Job type	Trigger (RecMem-inspired)
Merge candidate detection	Structural similarity across buffer fragments
Ontology promotion	Repeated reuse / $R(v)$ exceeds threshold
Symbolic abstraction	Equation transformation recurrence
Citation motif extraction	Graph density threshold in citation neighbourhood
Concept stabilisation	Cross-document persistence above $I(v)$ threshold
Decomposition repair	Topology inconsistency (sheaf gluing violation)

Table 6.9: CSP maintenance jobs with RecMem-inspired triggers, extending section 6.24.

- *Early phase* (sparse corpus): low $I(v)$ threshold for promotion; many tentative nodes; high tolerance for ambiguity.
- *Intermediate phase*: thresholds rise; merge candidates are resolved; graph topology stabilises around hubs.
- *Late phase* (mature ontology): high $I(v)$ threshold; only genuinely novel insertions trigger promotion; redundant fragments are garbage-collected.

RecMem's observation that memory consolidation exhibits *phase transitions* connects to this picture: the promotion of a concept from buffer to canonical node can be modelled as a first-order phase transition in the $I(v)$ landscape, with the annealing schedule determining the critical threshold. This opens a research direction in *graph cooling schedules* for semantic corpora, analogous to simulated annealing in combinatorial optimisation but operating over typed relation graphs rather than assignment spaces. The CSP energy function E_{total} (eq. (6.27)) provides a natural thermodynamic interpretation: semantic annealing minimises E_{total} with a temperature schedule $T(t) \to 0$.

Limits of RecMem for scientific cognition. RecMem lacks the representational substrate required for mathematical knowledge: no symbolic persistence across sessions, no formal graph morphisms, no algebraic compositional structure, no support for proof systems or equation lineage. A theorem that appears once but is proved by Lean is *certain*; a statement that recurs ten times but is never formalised is *speculative*. RecMem cannot distinguish these cases; CSP's Layer C (`cas-verified`, `l4:` fields) encodes exactly this distinction. Similarly, RecMem's vector clustering cannot recover the identity of Gauss's divergence theorem cited under fifteen different notation regimes; CSP's identity-preserving transclusion (section 6.26) handles this by construction.

Remark 6.4 (RecMem as a useful heuristic, not a semantic foundation). RecMem is an operationally clever and economical solution to a real engineering problem—conversational memory compression. The CSP architecture borrows its *asynchronous, trigger-based consolidation* and *delayed stabilisation*, but replaces embedding similarity with graph morphism compatibility and recurrence with the composite importance metric (6.28). This allows CSP to handle once-critical scientific statements—a unique theorem, a single counterexample, a definition that organises an entire field—that RecMem's frequency model would discard or fail to elevate.

7
Conclusion

The conceptual-spaces framework offers a geometrically grounded theory of cognition that occupies the middle ground between the rigidity of symbolic logic and the opacity of connectionist networks. Its core commitments—that quality dimensions are the primitive building blocks of meaning, that natural categories are convex regions, and that similarity reduces to distance—have generated a productive research programme spanning cognitive science, linguistics, robotics, and AI.

7.1 Summary of Main Results

Chapter 2 A conceptual space is a metric Cartesian product of quality domains. The weighted Minkowski metric captures the integral/separable distinction and provides the geometric substrate for all further analysis. The category-theoretic perspective supplies a complementary global structure. Moving beyond uniform metric spaces, *typed sparse vector spaces* generalise quality dimensions to heterogeneous, sparsely populated bases whose similarity functions are dispatched by type. Three enrichment layers (conceptual geometry, process semantics, type-theoretic grounding) jointly specify the full meaning of each basis element and enable detection of hidden novelty in scientific citations.

Chapter 3 Domains are groups of integral dimensions with their own internal metric and topology. Non-trivial topologies (e.g. circular hue) must be identified empirically. Hierarchical and non-linear dimensions—modelled by tree metrics and aligned via Gromov-Wasserstein distance—extend the framework to structured, discrete coordinate systems. Quality axes need not be hand-designed: the generative continuity criterion and decoder sensitivity metric provide principled tests for axes discovered by VAEs, while VQ-VAEs yield discrete CSP coordinates suited to categorical dimensions. Simplicial latent models extend coverage to multi-way citation dependencies.

Chapter 4 The convexity thesis—natural concepts are convex regions—is supported by cross-linguistic colour data, category-learning experiments, and the derivation of prototype effects as a theorem rather than a stipulation. Voronoi tessellations yield category boundaries consistent with convexity.

Chapter 5 Shepard's exponential law grounds similarity as inverse distance. Context-dependent weighting reconciles metric geometry with observed violations of symmetry and the triangle inequality. Structural importance weights (exponential decay with document depth, reference-based vocabulary scores) give a document-aware similarity function superior to token embeddings for structured scientific text. Multi-view similarity fuses textual, structural, and mathematical views into a single score learnable via perturbation-stability optimisation.

Chapter 6 Applications span colour naming, word embedding, robotics, knowledge representation, metaphor theory, large language models, mechanistic interpretability, theorem proving as CSP generation, epistemic compilation, computing system morphology, fractal constraint stacks, structured document processing, personal context modelling via conceptual-space morphisms, scientific semantic reconstruction via the Structure-from-Motion analogy (multi-view latent object recovery, semantic bundle adjustment, Gaussian semantic splatting, bibliography-as-sensor-fusion), CSP-to-CSP operators as a graph-enrichment algebra (Bayesian multi-view fusion, six operator families, semantic reconstruction loss), engineering systems as latent semantic objects (FMEA as failure-risk projection, hidden requirements as negative space, latent requirement excavation, temporal engineering semantics), the CSP energy landscape connecting energy-based models, VQ-VAE constraint codebooks, JEPA, and the layered reasoning stack of LLMs, EBMs, and formal verifiers, the ScholarWiki compact format (token-efficient tiddler serialisation, typed header grammar, transclusion syntax, semantic copy-paste via drag-and-drop), the event-driven semantic maintenance scheduler (event ontology, HFSM job families, nightly vacuum), the Equation Theories / Polymath case study demonstrating CSP as a collaborative mathematical infrastructure, and selective semantic crystallisation (RecMem reinterpretation: delayed consolidation, composite importance metric $I(v)$, multi-stage stabilisation pipeline, and adaptive semantic annealing).

7.2 Open Questions and Future Directions

Several important questions remain open:

1. **Learning the geometry.** How do agents acquire the metric structure of a conceptual space from experience? Unsupervised metric learning and Riemannian manifold methods are promising but have not yet been fully integrated with the CSP framework.

2. **Beyond convexity.** The convexity thesis fails for some human categories (disjunctive, chained, and goal-derived categories). Bechberger and Kühnberger [2017] formally demonstrate that convex regions cannot encode correlations between quality domains under

the canonical two-level metric (2.3) and propose fuzzy star-shaped sets (section 4.5) as a principled replacement. The full integration of star-shaped regions into the typed sparse framework (section 2.6) and the fiber-bundle CSP (section 2.15) requires specifying: (i) how the prototype parameter p^* is learned or inferred for each concept type; (ii) whether the star-shapedness condition (4.3) can be enforced as a differentiable regulariser in VQ-VAE training; and (iii) whether the α-cut family (4.4) provides a tractable parameterisation of graded category boundaries in high-dimensional typed sparse spaces.

3. **Dynamics and conceptual change.** Concepts change over time both within individuals (learning, forgetting) and across communities (semantic shift). A dynamic geometry in which the space itself deforms would capture these phenomena. Formally, the static coordinate $c \in \mathcal{M}$ should be replaced by a *temporal semantic trajectory* $c(t) \in \mathcal{M}(t)$, where the manifold itself evolves as new theorems impose new constraints. This enables detection of notation drift, paradigm bifurcation, and delayed recognition—the 4D analogue of the motion-field methods in recent dynamic scene reconstruction.

4. **Integration with neural architectures.** The relationship between conceptual-space geometry and the representational geometry of deep networks is partially understood but far from settled. Clarifying this relationship is essential for interpretability and for principled design of AI systems that must interact with humans.

5. **Personal and multi-agent spaces.** When multiple agents maintain private conceptual spaces layered over a shared public space, the system of morphisms $\{\phi_i\}$ forms a diagram in the category of conceptual spaces. Colimits of such diagrams represent *consensus spaces*—a formal model of shared understanding that could guide the design of collaborative AI systems.

6. **Product algebra for typed sparse spaces.** The typed sparse vector space of section 2.6 lacks a fully specified product operation. Candidates include a wedge-like exterior product (for nilpotent citations), a typed tensor product (for combining heterogeneous dimensions), and a graph composition operation (for path concatenation in citation graphs). Settling this algebra is a prerequisite for a rigorous category-theoretic treatment of typed conceptual spaces.

7. **Canonical similarity composition.** The three enrichment layers A, B, C each supply their own similarity function. How these should be combined—summed, weighted, integrated as a hierarchy, or composed via a learned mixing function—is unknown. The composition rule determines whether the overall similarity is a valid metric and whether it respects the geometric axioms of the CSP framework.

8. **Non-Euclidean latent geometry.** Current VAE and transformer representations embed documents in flat Euclidean spaces, but sci-

entific knowledge likely resides on a curved manifold with non-trivial topology: loops (circular definitions), branches (competing paradigms), quotient structures (equivalent formulations), and fiber bundles (parameterised families of theorems). Topological latent models—manifold VAEs, simplicial autoencoders, sheaf-theoretic representations—may capture this structure more faithfully.

9. **Semantic bundle adjustment.** The consistency optimisation of eq. (6.20) is stated as an objective function; its practical algorithm is unspecified. Sparse factor-graph methods (analogous to the Levenberg–Marquardt solver used in Structure-from-Motion), graph-neural energy minimisation, and Riemannian optimisation on the concept manifold are all candidates. Benchmarking these methods on a curated core of $O(10^3)$ mathematical papers with known formal dependencies (via Lean / Mathlib) would provide a concrete evaluation of the semantic reconstruction hypothesis.
10. **Engineering CSP evaluation.** The engineering CSP hypothesis of section 6.31—that FMEA tables, CAD models, firmware repositories, and field-failure logs are projections of one latent product graph—is stated but unvalidated. A concrete test would ingest all artefacts of a small real product, run the CSP→CSP operator pipeline, and measure whether latent-requirement excavation (section 6.31) correctly identifies constraints that caused historical field failures. The chromium-to- stainless steel substitution failure is an ideal initial case study.
11. **EBM training and calibration for CSP.** The EBM energy functions of eq. (6.27) are currently abstract. Concrete learning algorithms are needed: how are λ_i and E_{ij} trained from a corpus of partially annotated CSP graphs? What loss function drives the VQ-VAE constraint codebook toward semantically meaningful clusters? How does the JEPA prediction objective combine with the EBM consistency objective during joint training? Specifying these training procedures, and benchmarking the layered LLM + EBM + Lean stack on formal reasoning corpora, would concretely evaluate the energy-landscape interpretation of CSP.
12. **Source calibration and Doppelgänger disambiguation.** Every bibliographic source introduces systematic biases analogous to camera calibration parameters: abstraction level, notation era, venue rigor, and author community. Estimating these "semantic camera parameters" jointly with the manifold reconstruction is the CSP analogue of calibrated multi-camera SfM. A related challenge is the Doppelgänger problem: distinct concepts that appear symbolically identical (overloaded notation, copied derivations, same theorem under different names). Resolving these requires provenance, citation neighbourhood, and theorem-dependency context—dimensions that purely token-similarity systems cannot access.
13. **Semantic digestion as an optimisation target.** The Semantic Digestion Score SDS of eq. (2.10) is proposed as a metric but

not yet as a trainable objective. How should SDS be learned from human feedback or proxy signals (downstream reuse count, pedagogical projection quality, student comprehension ratings)? What loss function over the CSP graph turns Tao's implicit mathematical goals [Tao, 2026]—exposition, insight transfer, teachability—into differentiable constraints on the semantic compiler?

14. **Event-driven semantic maintenance at scale.** The maintenance scheduler of section 6.24 specifies job families and triggers at the design level. The open questions are operational: what is the optimal scheduling policy for background jobs (similarity detection, abstraction induction, contradiction resolution) in a graph of $O(10^6)$ nodes? How to balance responsiveness (low-latency `CONTRADICTION_FOUND` resolution) against computational cost (nightly semantic vacuum)? Can the HFSM scheduler be formally verified for liveness and deadlock freedom?

15. **Multi-objective formalisation.** The typed sparse vector space currently encodes logical and structural dimensions. Extending it with explicit fields for *pedagogical level* (introductory/intermediate/research), *modularity* (self-contained vs. deeply dependent), and *expository quality* (narrative coherence score) would allow retrieval to filter on these dimensions directly. This would operationalise Tao's "separate workflows" recommendation: a fast, correctness-only track and a slow, digestion-optimised track within the same CSP graph, differing only in which fields are populated and at what fidelity.

16. **Proof-to-talk transform.** Tao's "can someone give a talk?" criterion becomes a concrete compiler pass: a transform that takes a formal Lean proof and generates a slide skeleton, a spoken narrative, or an interactive tutorial. The inputs are the Lean proof term (`l4:` field), the dependency DAG (`f:` field), and the Layer A conceptual neighbourhood. The output is a $\texttt{teach}(c) = 1$ tiddler with a SLIDE_DECK or SPOKEN_MATH projection. Developing and evaluating this transform on the Equation Theories corpus (section 6.29) would provide a direct empirical test of the Digestion Layer D proposal.

Despite these open questions, conceptual spaces remain one of the most coherent and empirically grounded frameworks for the geometry of meaning, and their influence on both cognitive science and AI continues to grow.

8
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